

Software Architecture

Exercise – Business Development

BSc



Ingo Arnold

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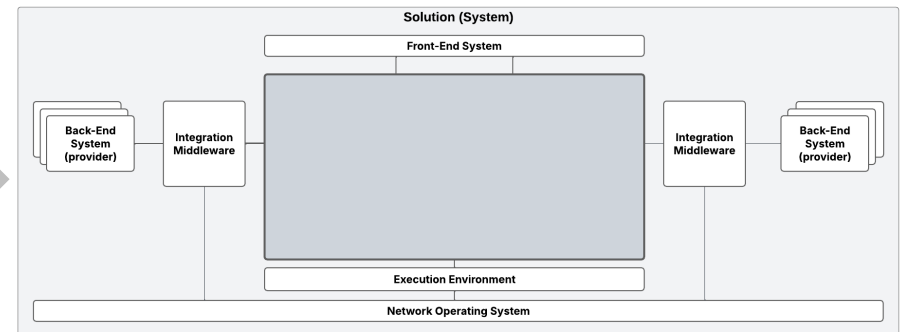
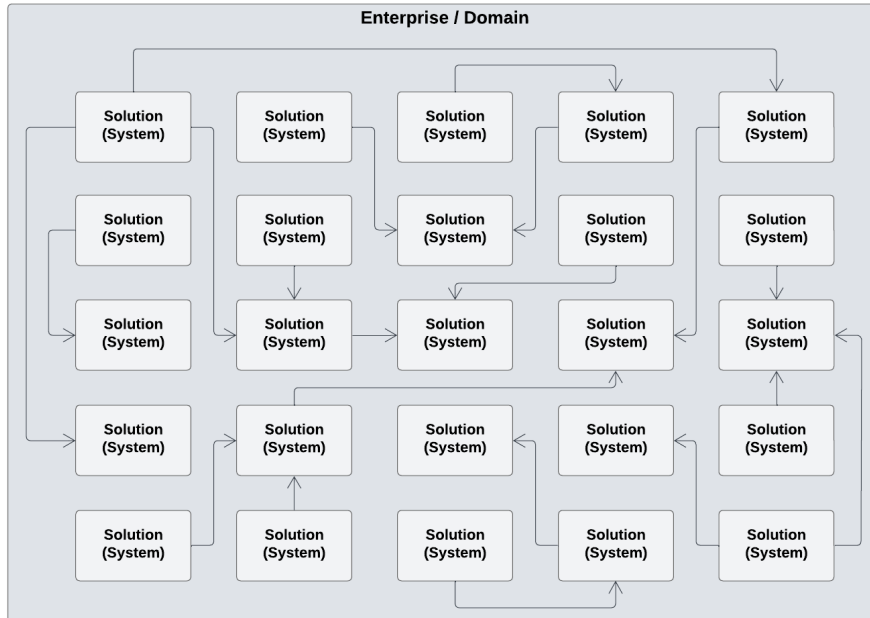
Exercise Opening

Overview

In **today's exercise**, we will look at the **interaction** between the **Architecture Disciplines** of Enterprise, Domain, and Solution Architecture, and finally Software Architecture, using an **example** of **end-to-end** and **top-to-bottom architecture development**.

We will **discuss the case together** and **you will solve embedded tasks** and **answer questions** along the way.

We start at the **Enterprise/Domain Level** and then **zoom in to the level of an individual Solution ...**

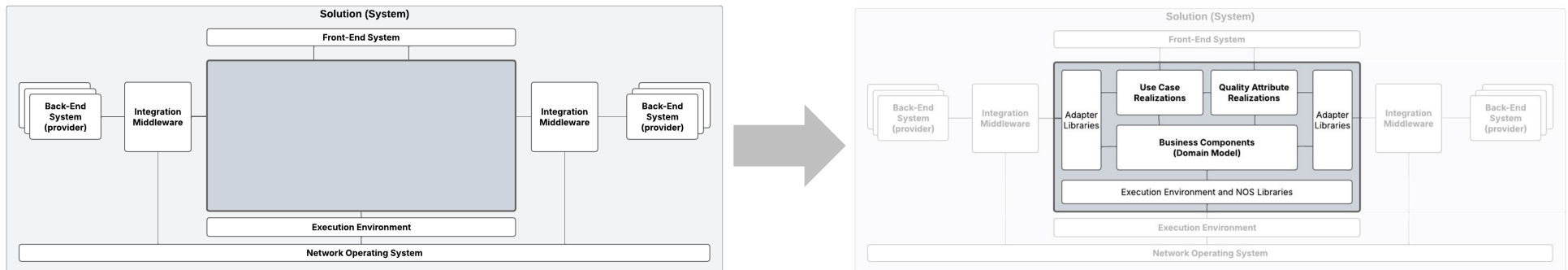


Exercise Opening Overview

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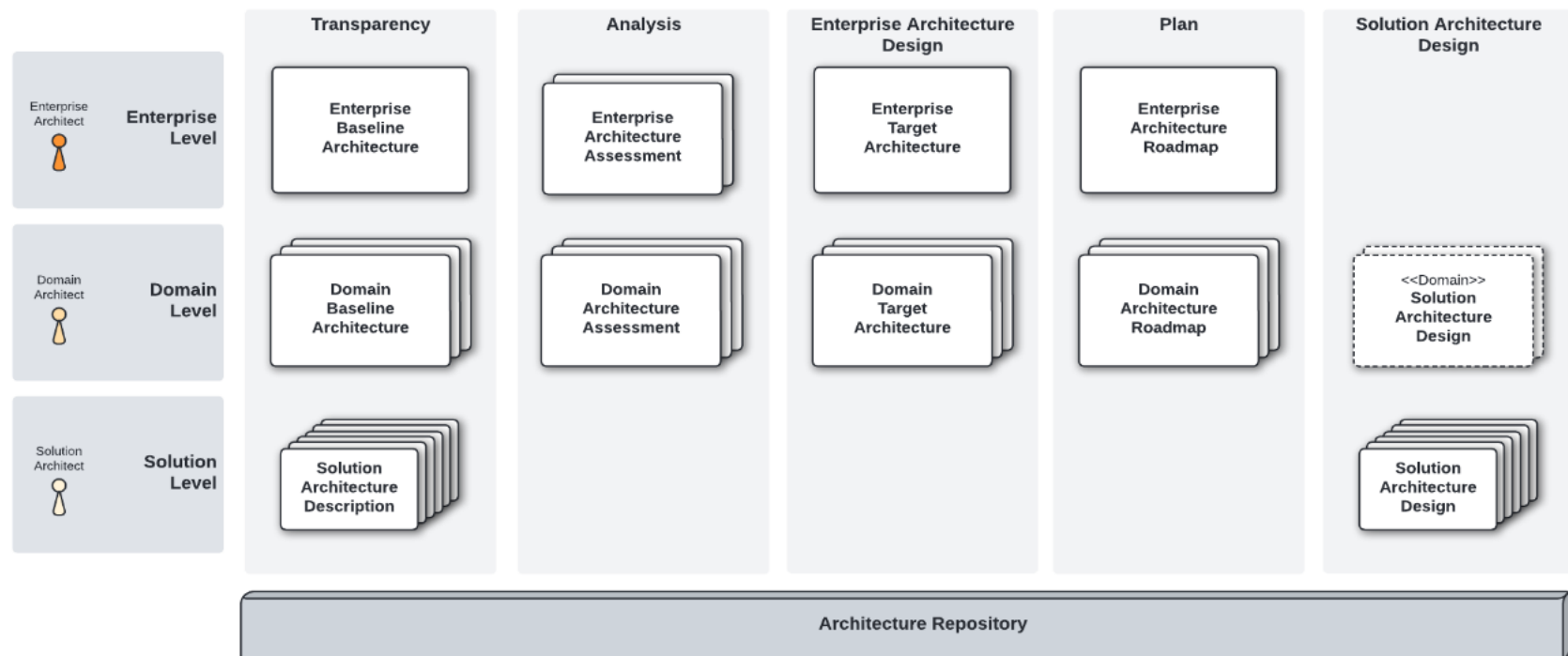
... and then **zoom in even further** to the **responsibilities** of the **Software Architect**.



Exercise Opening

Overview

The **case** we are looking at is **based** on the **fictional bank** you encountered in Deck [5.1-BSc_SWA_HOW_Method.pptx](#) and also **follows** the **scheme** introduced there.



Case Study

Overview

The **Bankolino Enterprise Architect (EA)**, together with the responsible **Domain Architect (DA)**, is tasked with **investigating** the **advantages** and **disadvantages** of completely **outsourcing personnel recruitment** to a suitable **BPO partner** (BPO = Business Process Outsourcing) for Bankolino.

Currently, Bankolino's **entire recruitment and talent acquisition process** is **operated based on its own application portfolio**.

Bankolino's management is **considering outsourcing** this **process** to a **suitable BPO partner**, since it is a **non-strategic area** of the bank's capabilities.

Case Study

Overview

What does **BPO** (**B**usiness **P**rocess **O**utourcing) mean?

Case Study Overview

What are the **advantages** and **disadvantages** or risks of **BPO** projects?

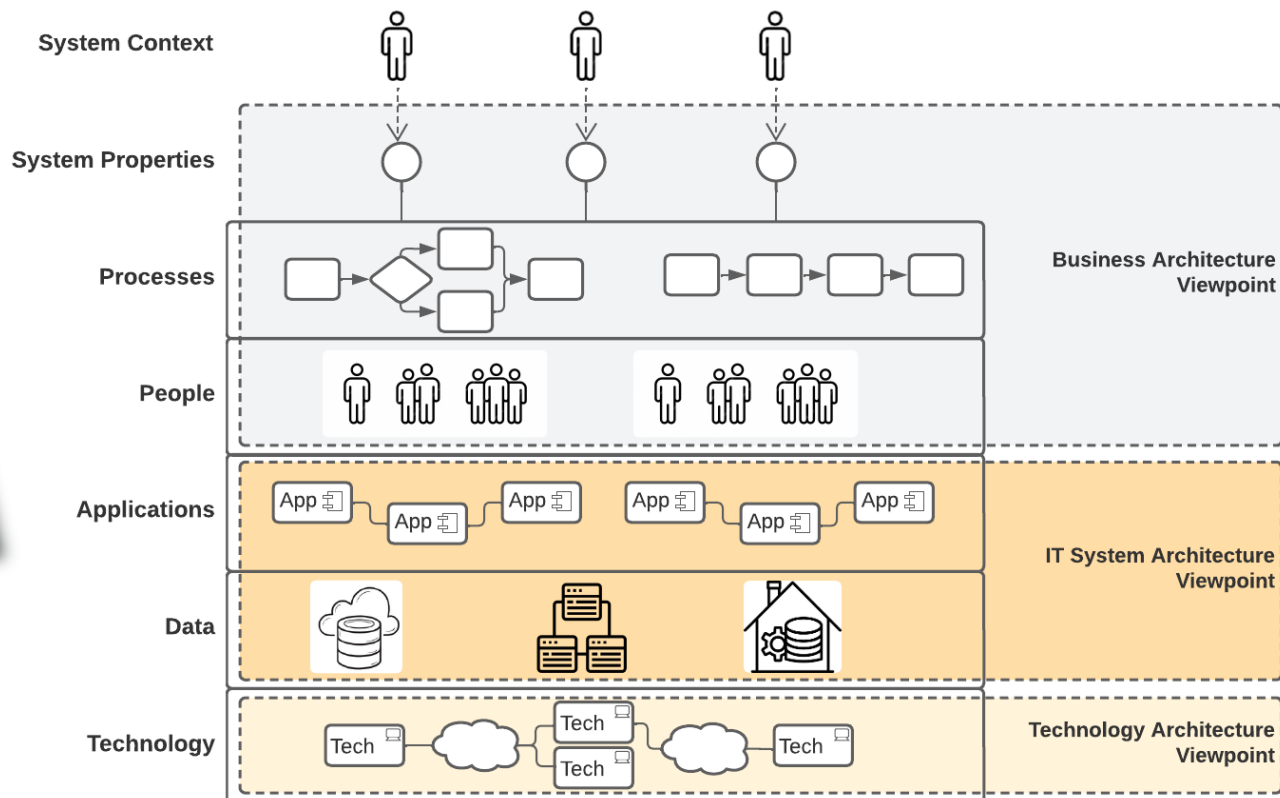
Enterprise & Domain Architecture

Holistic View of an Enterprise & Domain

Enterprise and Domain Architects view the various (types of) Building blocks of an Enterprise (socio-technical system) and their interrelationships, holistically.

Different **types of Building Blocks** require quite **different approaches** to their elaboration.

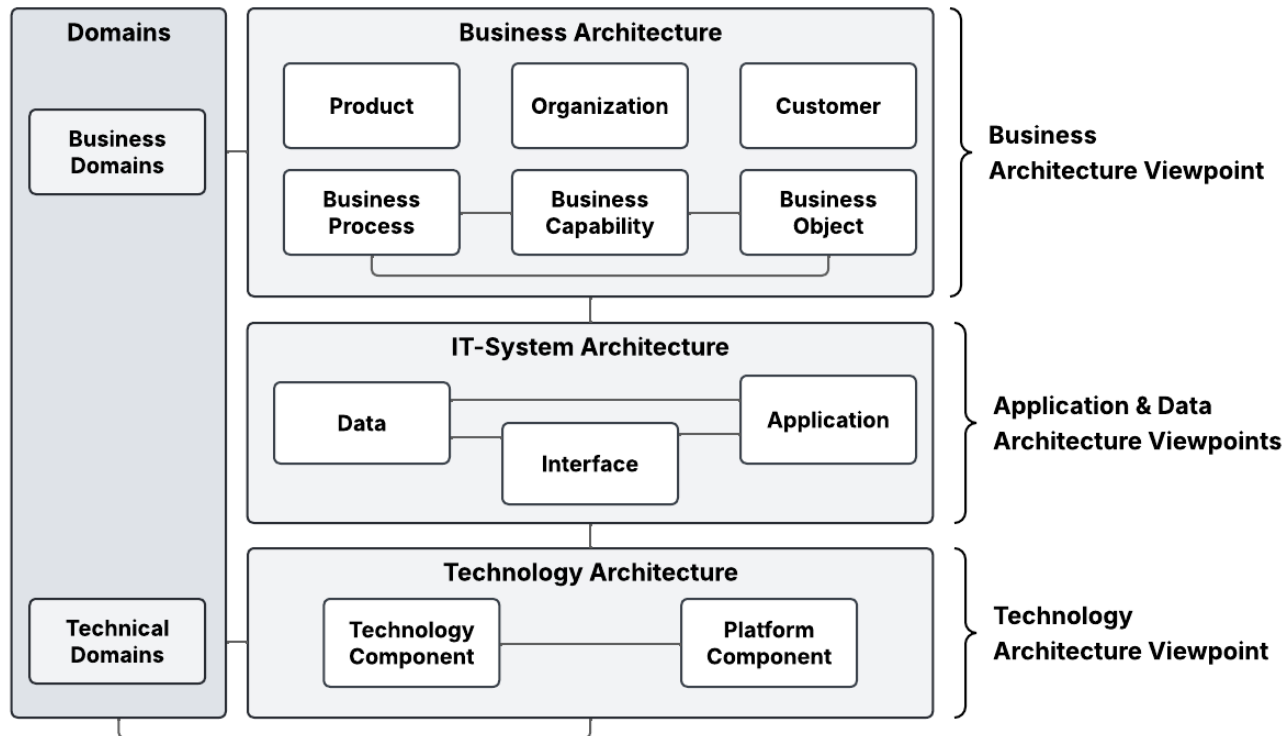
Hence, different **Enterprise Architecture Viewpoints** have emerged that **address particular types of Building Blocks**.



Enterprise & Domain Architecture

Holistic View of an Enterprise & Domain

Each of the **Enterprise Architecture** viewpoints captures **Building Blocks** of a **specific type** and **respective** (types of) **relationships** between them.

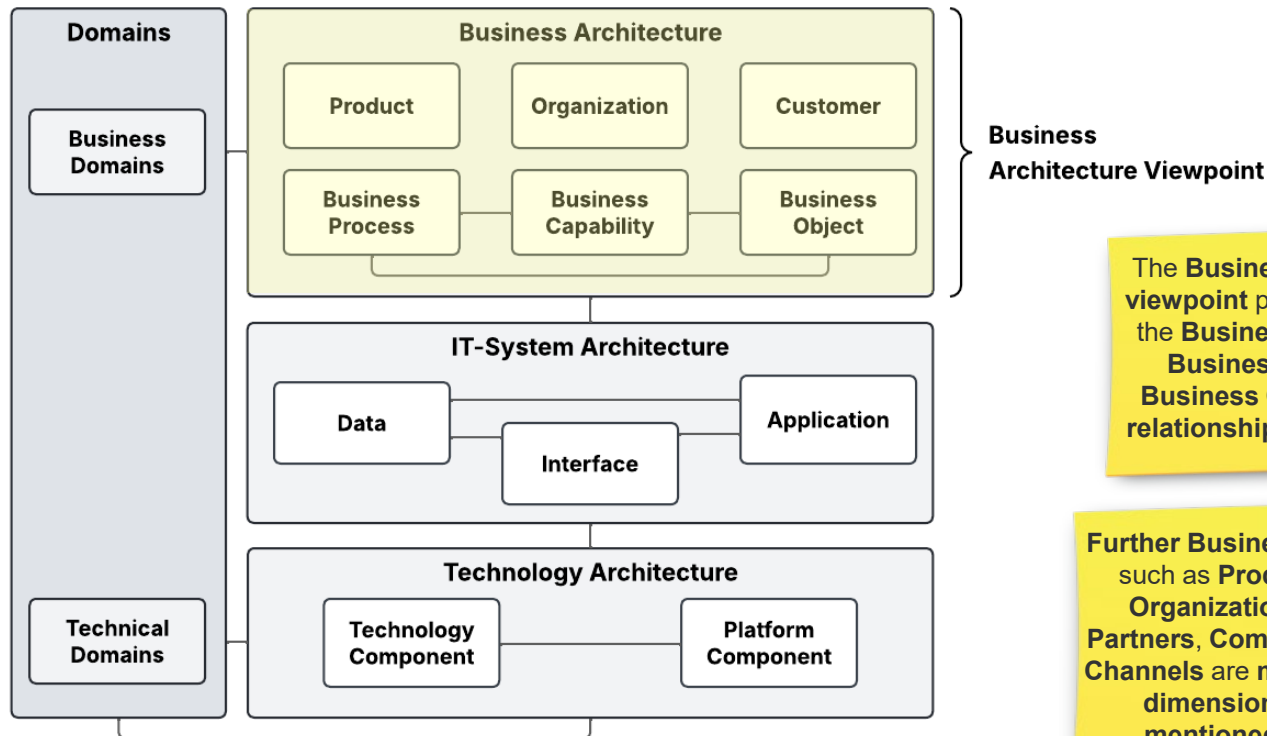


(high-level)
**Enterprise
Architecture
Metamodel**

Enterprise & Domain Architecture

Holistic View of an Enterprise & Domain

Each of the **Enterprise Architecture** viewpoints captures **Building Blocks** of a **specific type** and **respective** (types of) **relationships** between them.



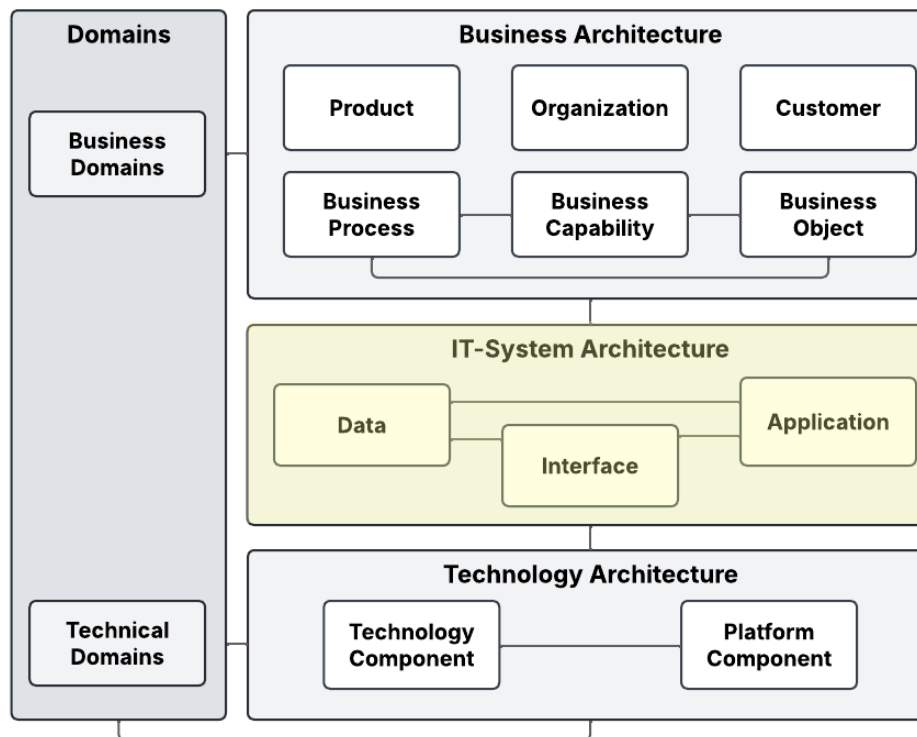
The **Business Architecture** viewpoint primarily captures the **Business Capabilities**, **Business Processes**, **Business Objects** and the **relationships** between them.

Further **Business Building Blocks** such as **Products & Services**, **Organization**, **Customers & Partners**, **Communication & Sales Channels** are mapped as **business dimensions** to the above-mentioned **Components**.

Enterprise & Domain Architecture

Holistic View of an Enterprise & Domain

Each of the **Enterprise Architecture** viewpoints captures **Building Blocks** of a **specific type** and **respective** (types of) **relationships** between them.



The **Application Architecture Viewpoint** captures **Applications, Data, and Interfaces** as well as **relationships** all among these.

Applications offer their users **valuable functionalities** ...

Application & Data Architecture Viewpoints

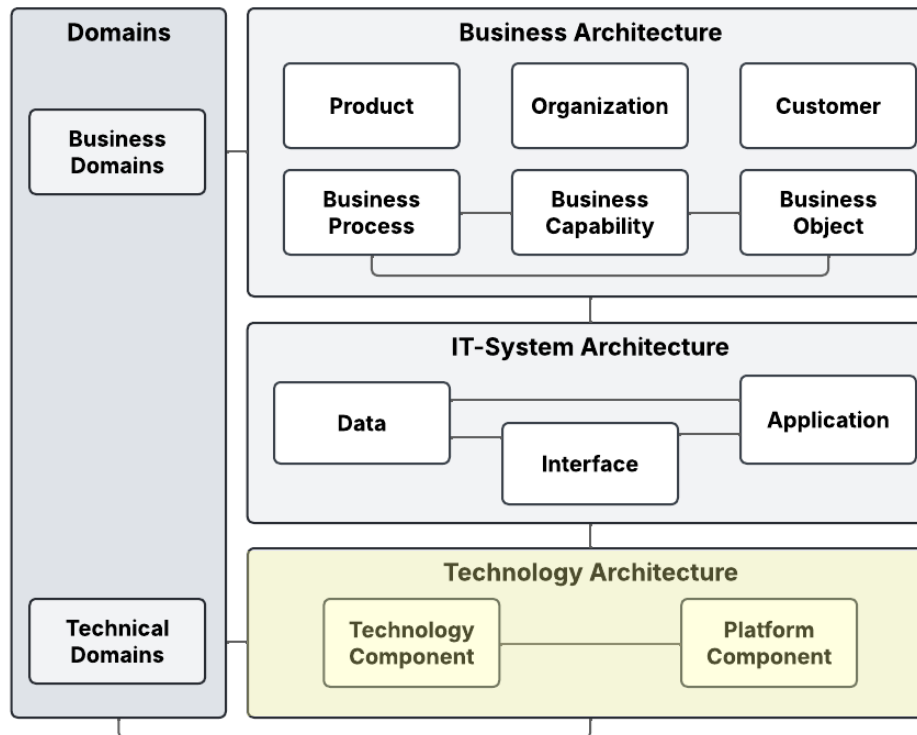
... and are **connected** to other applications via **interfaces**.

Further, **Applications** encapsulate **Data**.

Enterprise & Domain Architecture

Holistic View of an Enterprise & Domain

Each of the **Enterprise Architecture** viewpoints captures **Building Blocks** of a **specific type** and **respective** (types of) **relationships** between them.



Technology Components are the **technical products** or **services** that a company purchases from **vendors** in their **COTS** state.

The **Technology Architecture Viewpoint** captures **Technology** and **Platform components** and the **relationships** between them.

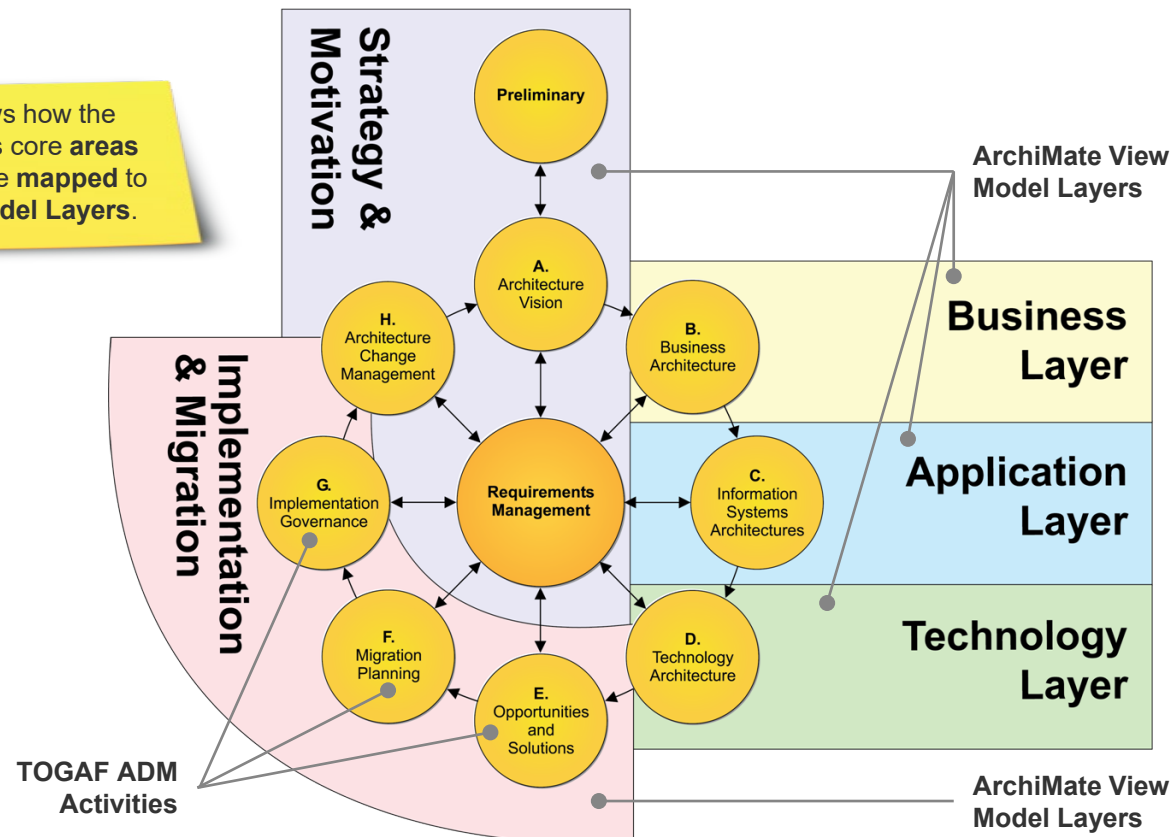
Platform Components are the **building blocks** that a company **instantiates** based on **Technology** and are hence **customised** to the **needs** of the **company**.

Enterprise & Domain Architecture

Holistic View of an Enterprise & Domain

ArchiMate and **TOGAF** are both **Open Group** standards. While **TOGAF** focuses on **process (ADM)**, **ArchiMate** contributes the **View Model**.

This diagram shows how the **TOGAF ADM** and its core **areas of activity (A .. H)** are **mapped to ArchiMate View Model Layers**.



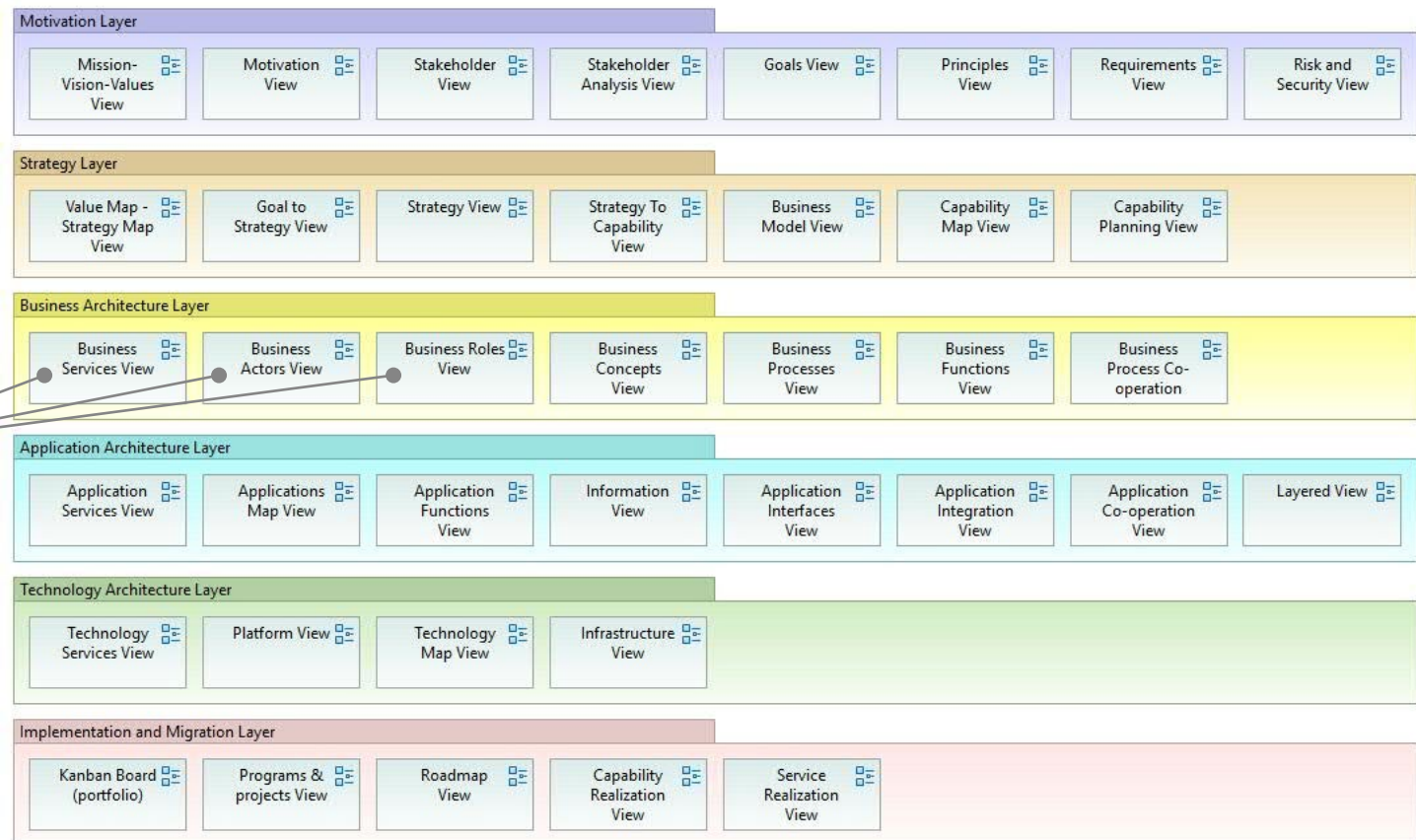
Enterprise & Domain Architecture

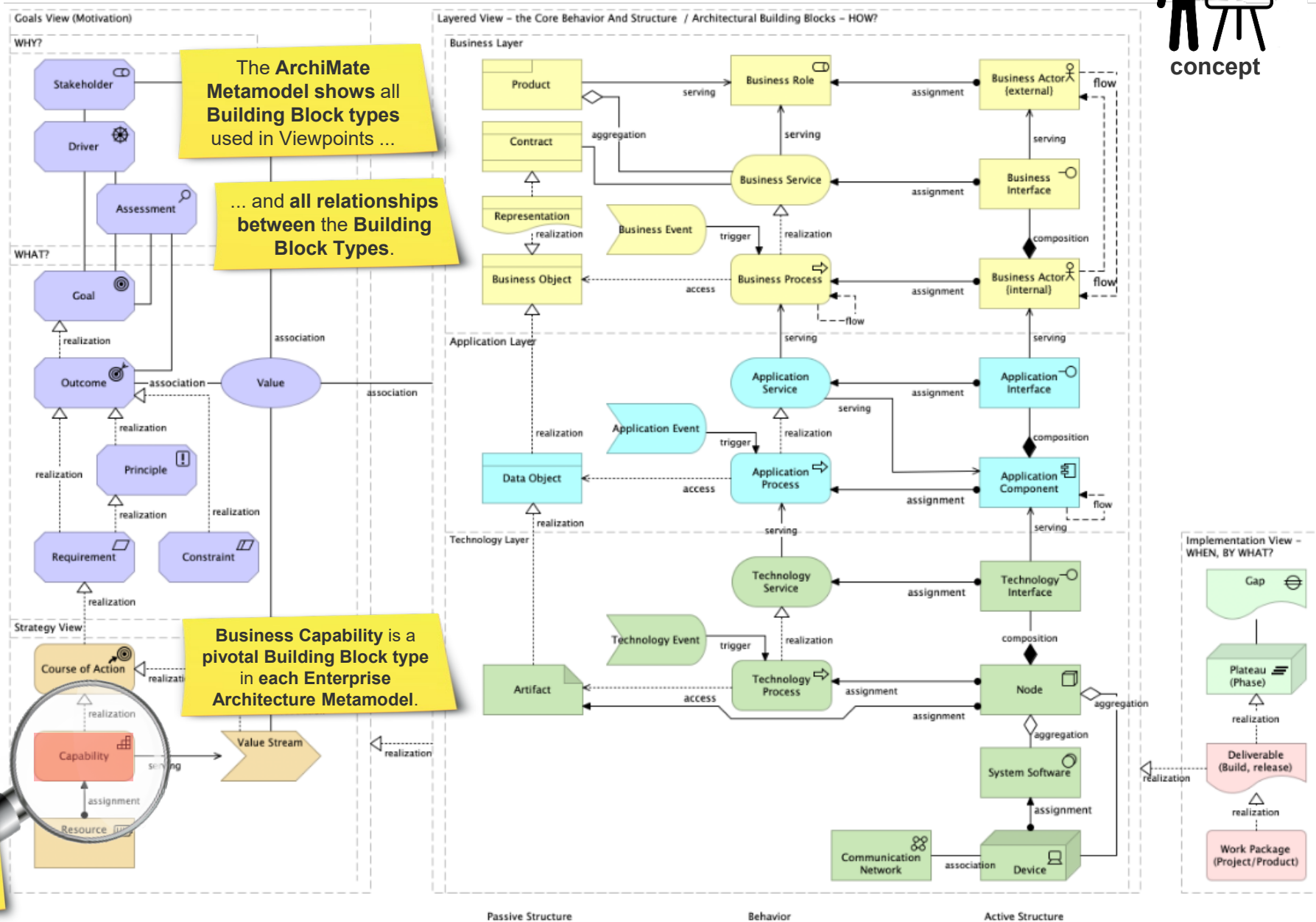
Holistic View of an Enterprise & Domain

Each of the **Enterprise Architecture viewpoints** captures **Building Blocks** of a **specific type** and **respective** (types of) **relationships** between them.

Each **ArchiMate View Model Layer** consists of a set of **Viewpoints** that offer interesting insights for **Enterprise** and **Domain Architects**.

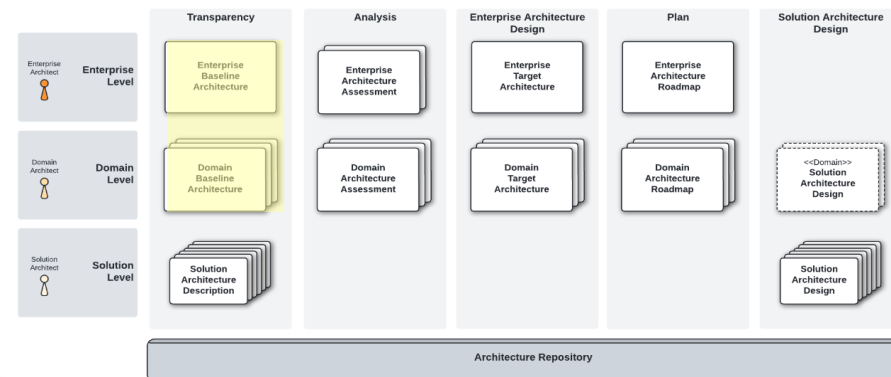
ArchiMate Viewpoints of the **Business Architecture Layer**





In a next step we start with a Business Capability Map to look into a Business Development scenario.

Case Study BPO Example



Key objective of **Business Capability Management** is to use knowledge of business structures and their relationships to IT structures in order to systematically identify and leverage consolidation and other improvement opportunities.

Domain

Business Capabilities

Retail Banking	Account Management	Customer Onboarding	Loan Processing	Mortgage Processing	Payment Processing	Savings Management	Credit Card Management	Personal Loans	Branch Operations
Wealth Management	Investment Advisory	Portfolio Management	Retirement Planning	Trust and Estate Management	Private Banking	Asset Allocation	Client Rel. Management	Tax Advisory	Investment Research
Risk Management	Credit Risk Management	Market Risk Management	Operational Risk Management	Compliance and Regulatory Risk	Liquidity Risk Management	Fraud Detection and Prevention	Risk Reporting and Analytics	Scenario Analysis	Stress Testing
IT and Digitalization	Core Banking Systems	Digital Banking Platforms	Data Management	Cybersecurity	IT Infrastructure	IT Support and Operations	Application Development	Cloud Computing	IT Governance
Marketing and Communication	Customer Segmentation	Campaign Management	Social Media Engagement	Brand Management	Public Relations	Customer Analytics	Market Research	Digital Marketing	Customer Feedback Mgmt.
Operations and Backoffice	Transaction Processing	Settlements and Clearing	Reconciliation	Cash Management	Document Management	Supplier Management	Call Center Operations	Trade Finance Operations	Collateral Management
Compliance and Legal	Regulatory Compliance	Anti-Money Laundering	Know Your Customer	Legal Risk Management	Contract Management	Audit Management	Policy Management	Sanctions Monitoring	Legal Advisory
Human Resources	Recruitment and Talent Acquisition	Learning and Development	Performance Management	Payroll Management	Compensation and Benefits	Workforce Planning	Employee Relations	Health and Safety	Diversity and Inclusion



In a next step the **Enterprise Architect** zooms into the **Recruitment and Talent Acquisition** capability to better understand its next level of detail capabilities.

The **Enterprise Architect** differentiates business capabilities in terms of the importance with which **Bankolino** keeps their strategic and operational control in-house versus delegating it to an external party.

Blue: Fully internal capabilities that are critical to the strategic and operational steering of the bank.

Purple: Mixed sourcing, partly internal and partly outsourced.

Grey: Capabilities that can be fully outsourced (i.e., non-core activities).

Case Study

Overview

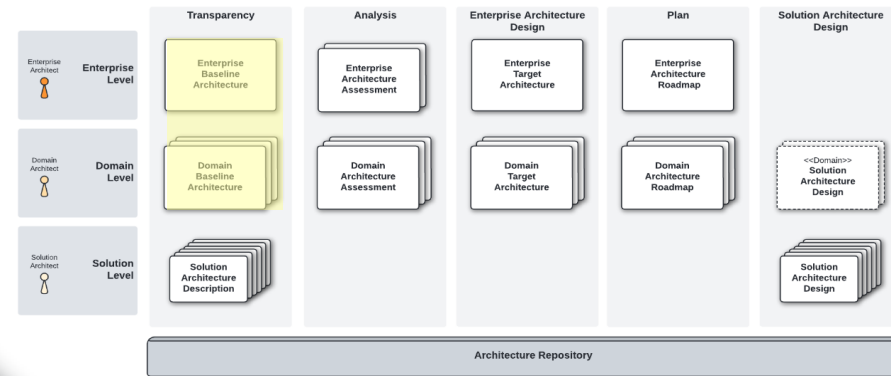
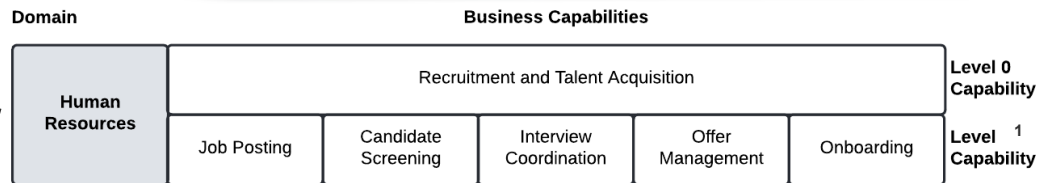
What **further aspect** do we **consider** in the **next step** and is **associated** with the **capabilities** we just started with?

- 1st | Processes
- 2nd | Applications
- 3rd | Information and Data
- 4th | Organization
- 5th | Services
- 6th | Platforms
- 7th | Technology

Case Study BPO Example



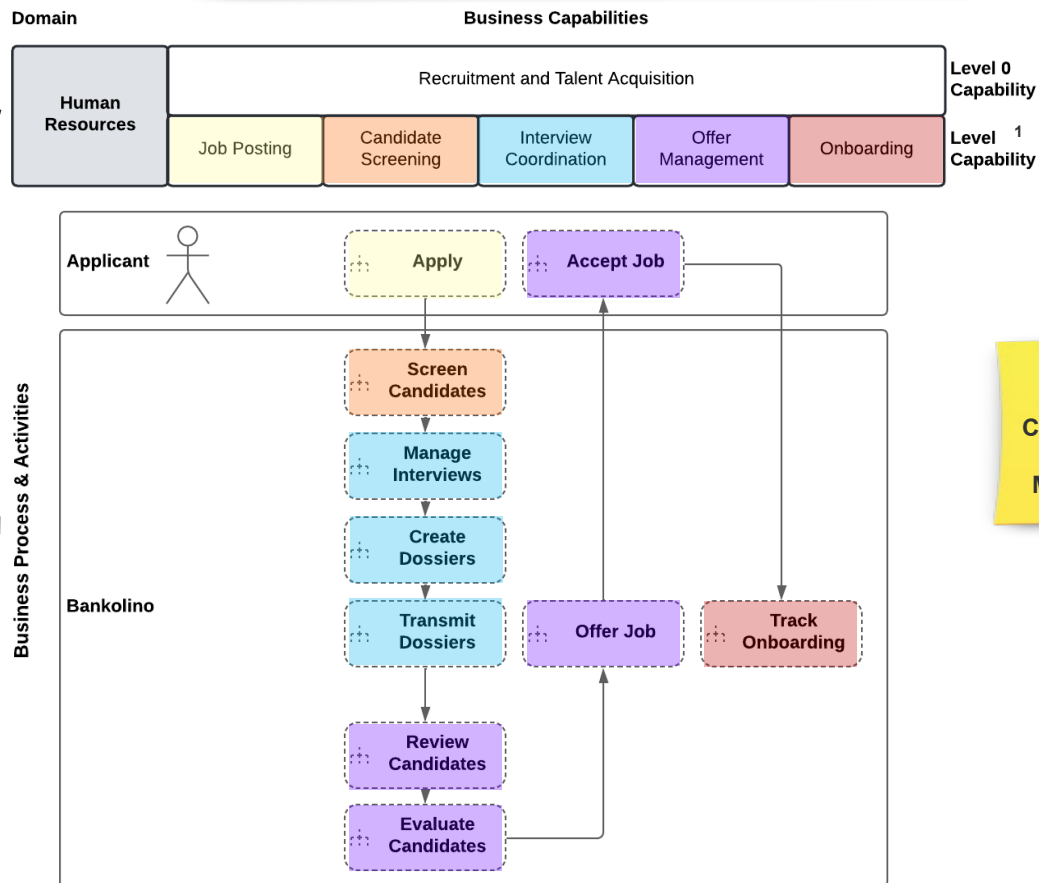
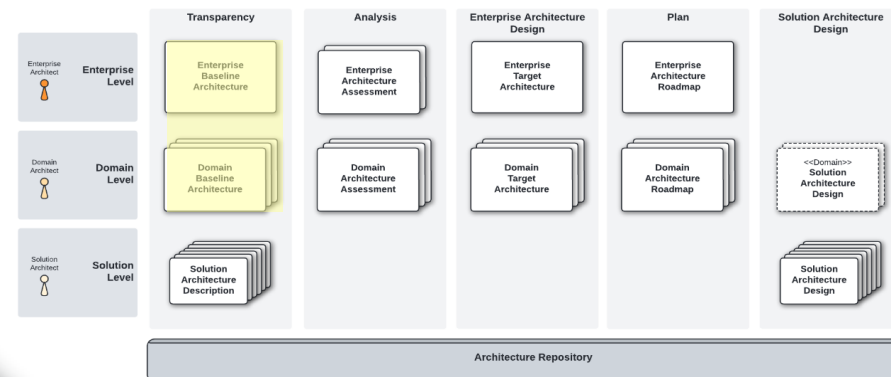
Business capabilities and **processes** are **both associated** with **activities** at the **business model level** (Business Model Canvas), which is why there is a **close relationship between these two concepts**.



Case Study BPO Example



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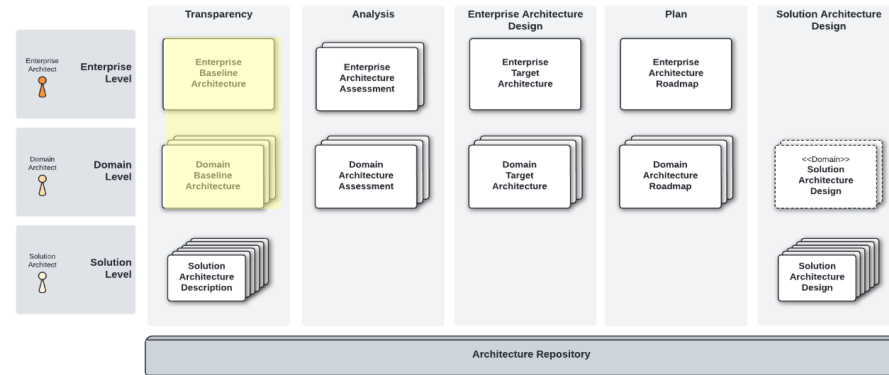


However, while business capabilities support a static view of the abilities that a business fundamentally needs ('what'), business processes tend to support an operational perspective ("how" and 'who').

Enterprise and Domain Architects review the detailed Business Capabilities and the associated process map to understand the scope of the Management-requested investigation.

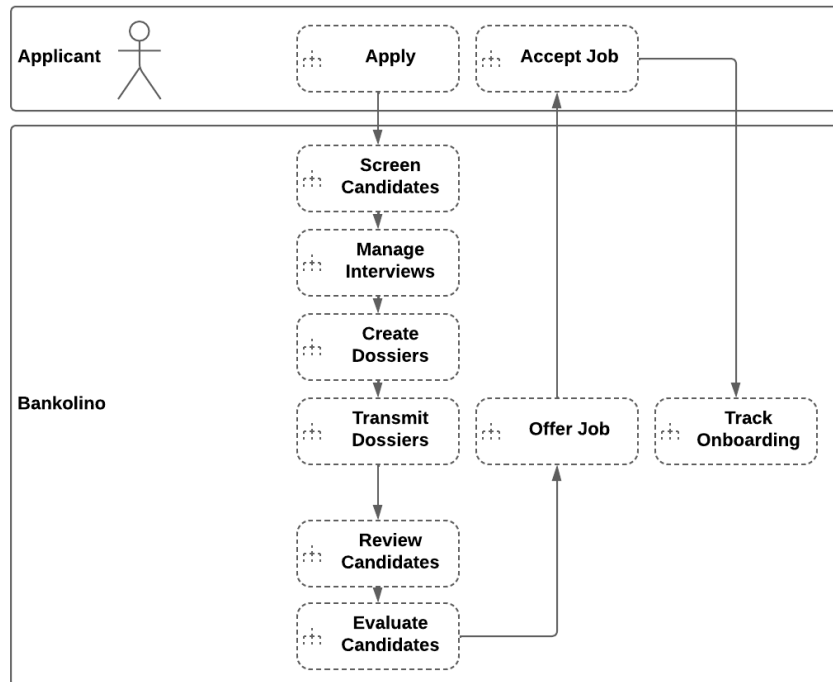
They recognize that all process steps are currently operated by Bankolino.

Case Study BPO Example

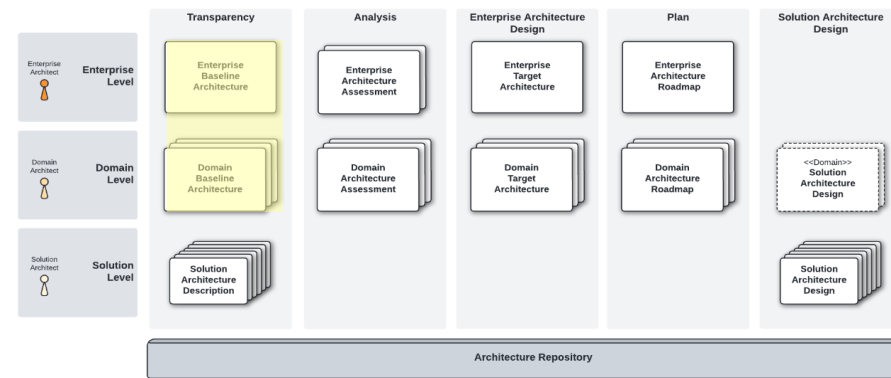


Business Processes, and their **corresponding activities**, are in turn (partially) **supported** by **applications** that **provide functions** and **make processes particularly efficient and effective**.

Business Process & Activities



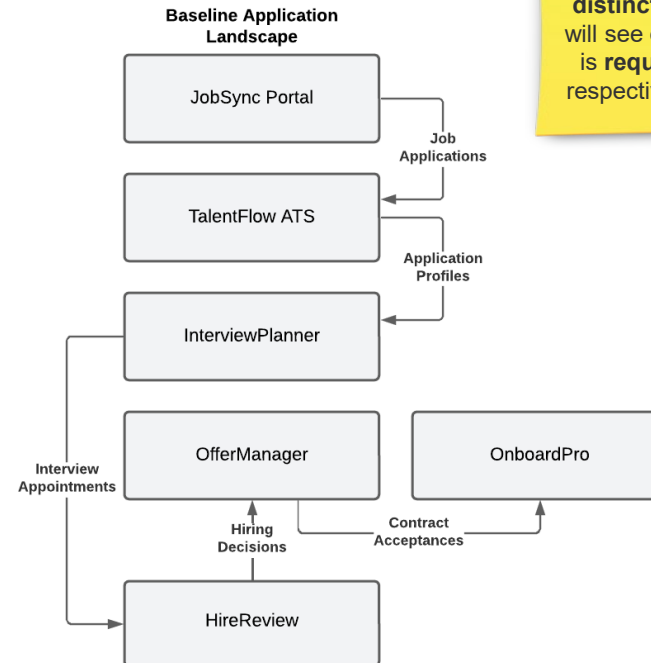
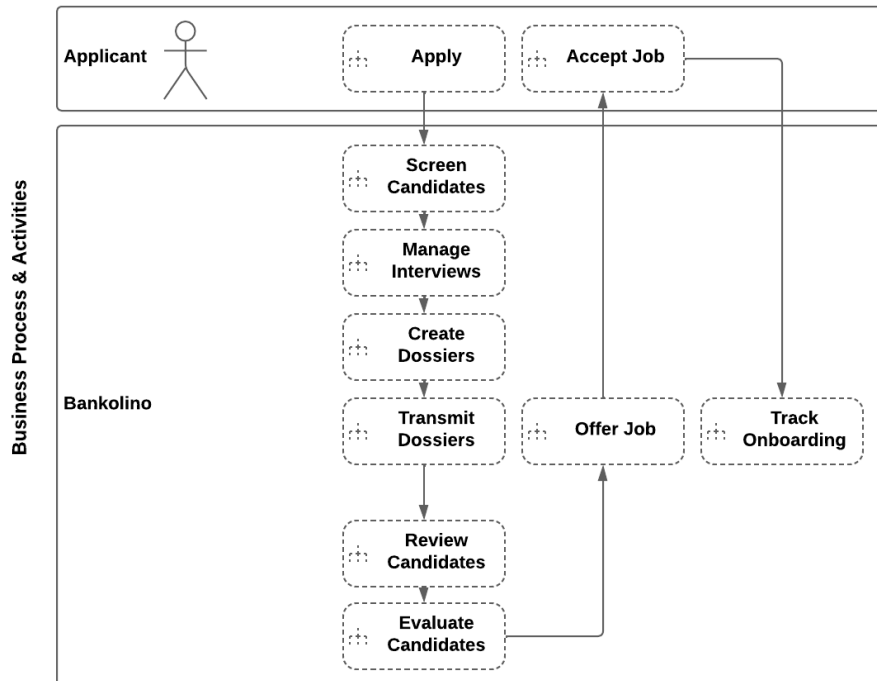
Case Study BPO Example



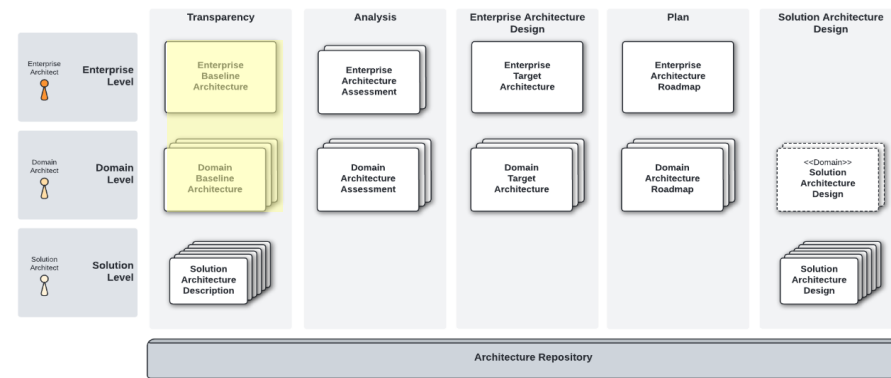
Business Processes, and their **corresponding activities**, are in turn (partially) **supported** by **applications** that **provide functions** and **make processes particularly efficient and effective**.

The **applications exchange information objects** and are, hence, **heavily interconnected**.

Each **application offers distinct functionality** (as we will see on the next slides) that is **required to underpin** the respective **process activities**.

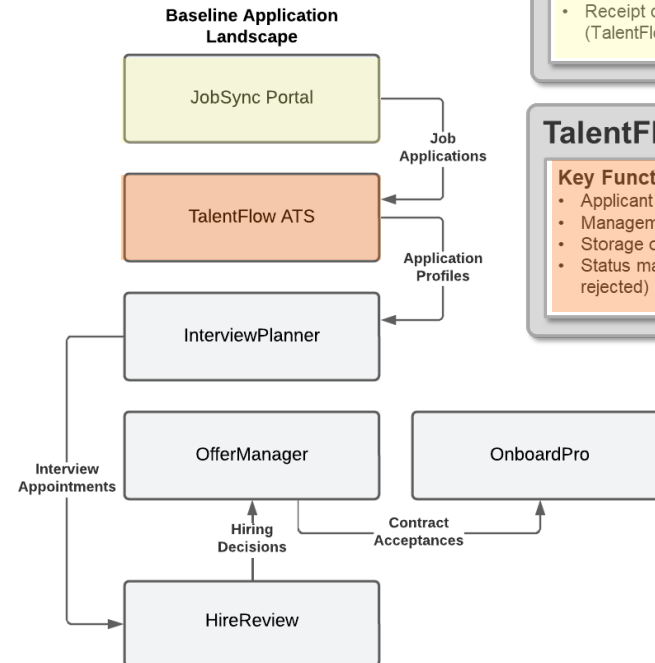
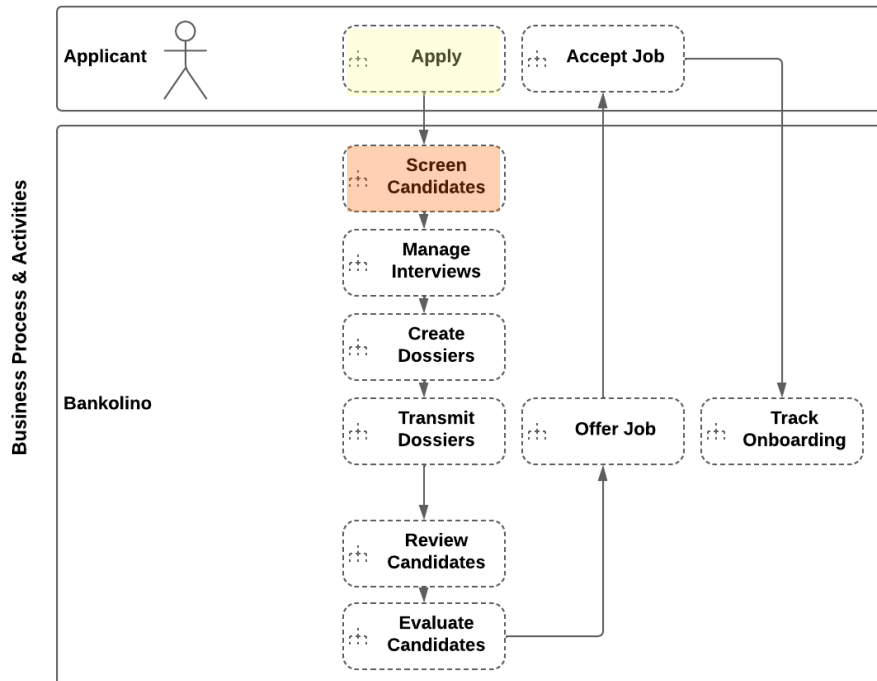


Case Study BPO Example



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JobSync Portal

Key Functionality

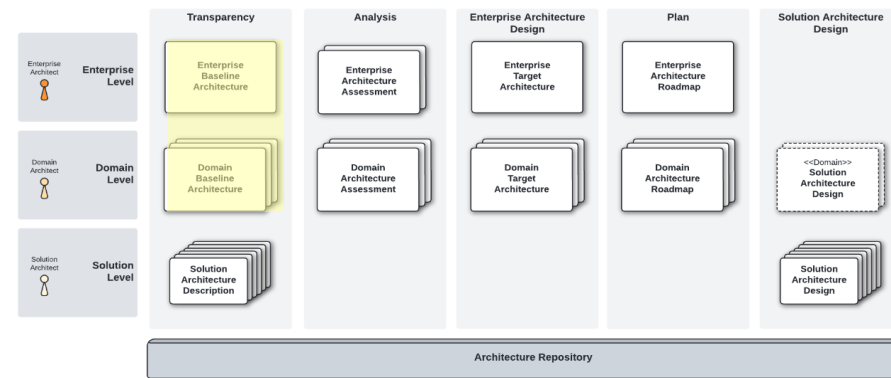
- Creation and publication of job advertisements on external job boards.
- Management of job postings by location/department.
- Receipt of applications directly into the ATS (TalentFlow)

TalentFlow ATS

Key Functionality

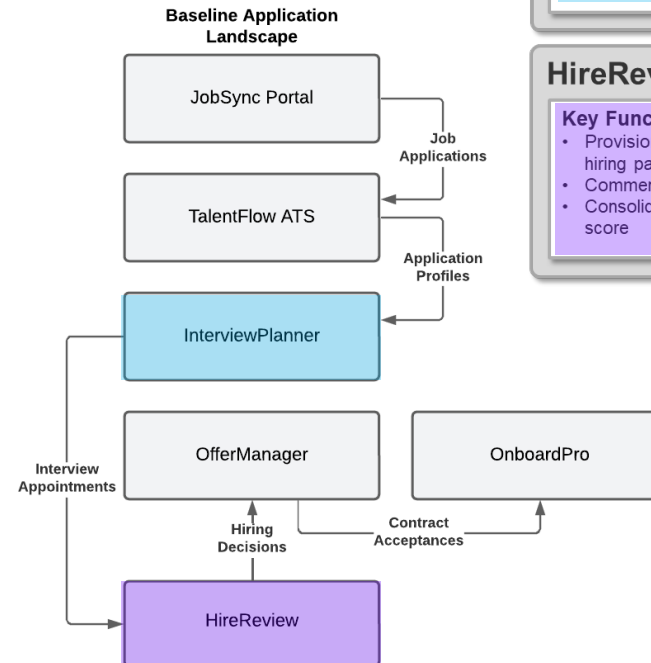
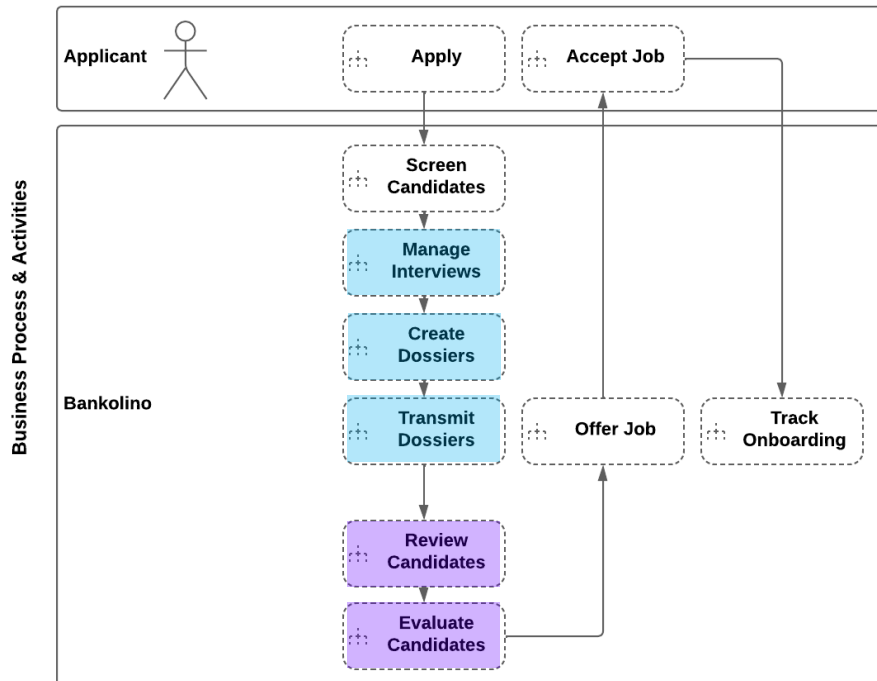
- Applicant Tracking System.
- Management of incoming applications
- Storage of applicant data and documents.
- Status management (e.g. received, in progress, rejected)

Case Study BPO Example



Business Processes, and their **corresponding activities**, are in turn (partially) **supported** by **applications** that **provide functions** and **make processes** particularly efficient and effective.

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InterviewPlanner

Key Functionality

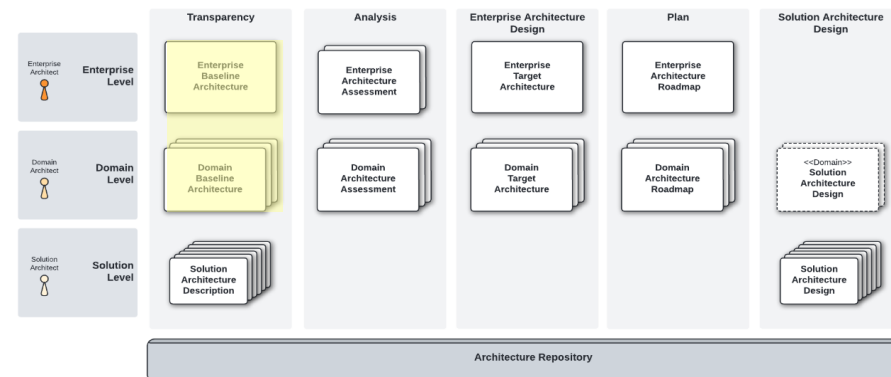
- Scheduling job interviews (automated & manual).
- Calendar integration with internal employees.
- Sending appointments & reminders to applicants

HireReview

Key Functionality

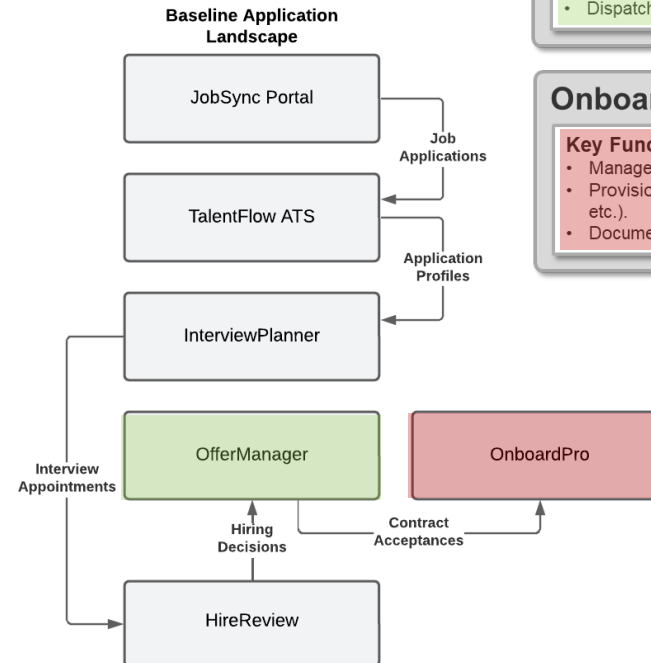
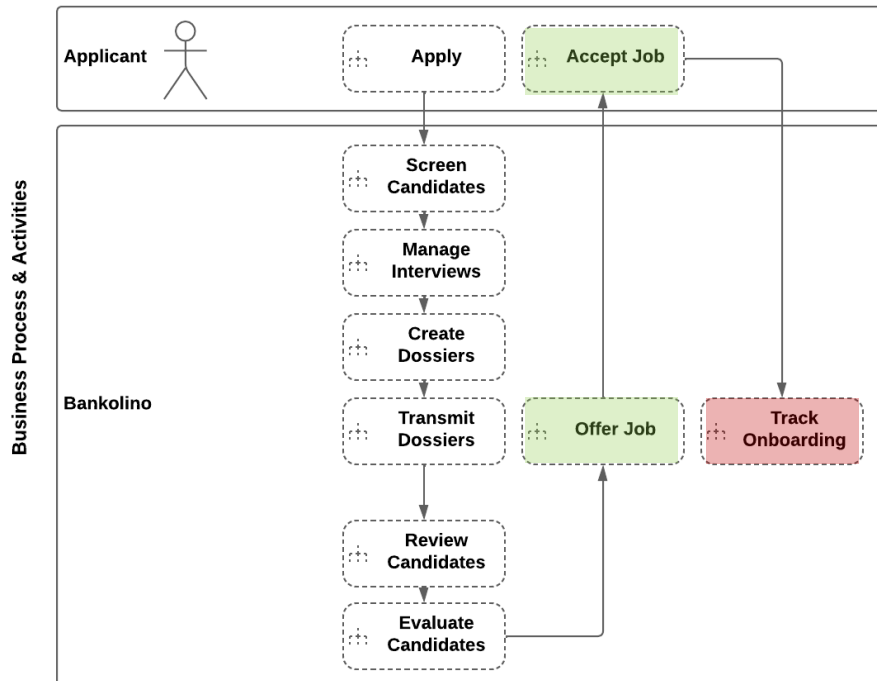
- Provision of structured evaluation templates for hiring panels.
- Comment and note function.
- Consolidation of evaluations and derivation of a score

Case Study BPO Example



Business Processes, and their **corresponding activities**, are in turn (partially) supported by applications that **provide functions** and make processes particularly efficient and effective.

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OfferManager

Key Functionality

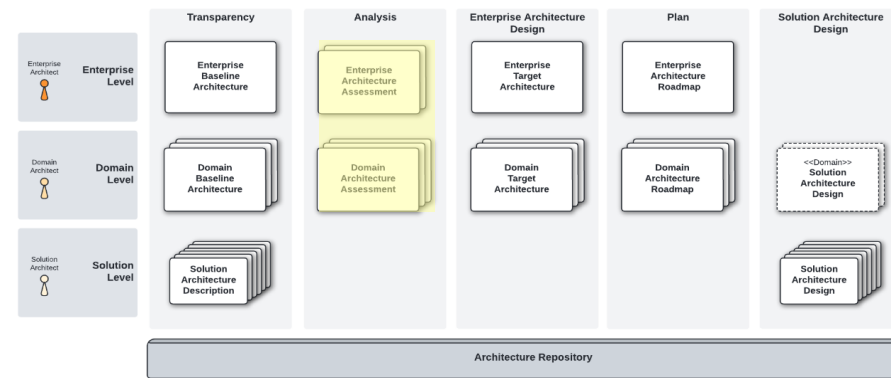
- Creation of contract offers based on templates.
- Approval workflows for HR and specialist departments.
- Dispatch and tracking of offer acceptances

OnboardPro

Key Functionality

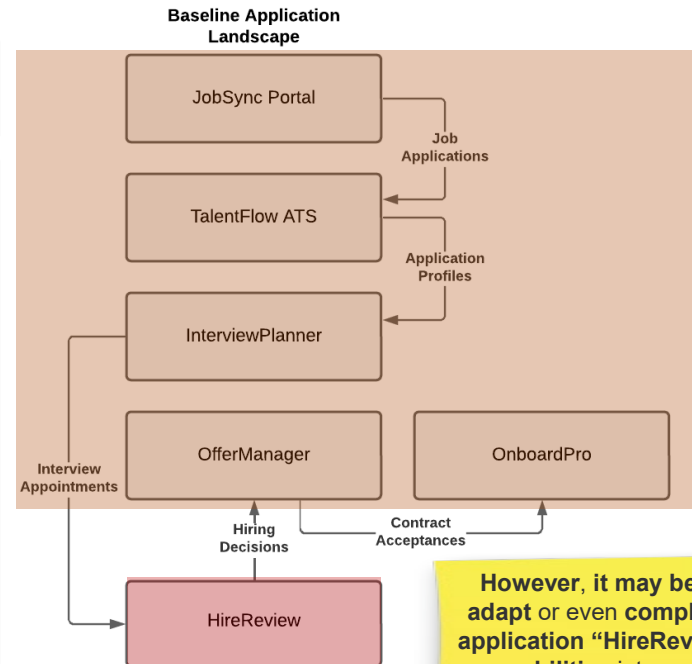
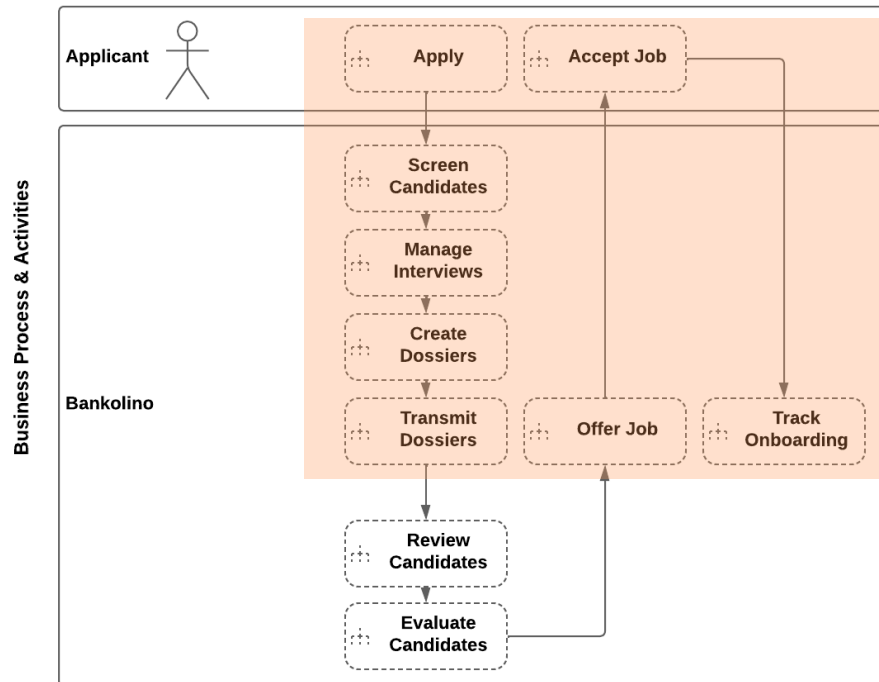
- Management of the onboarding process.
- Provision of task lists (upload contract, IT access, etc.).
- Document management & compliance checks

Case Study BPO Example



Next the **architects move into the Analysis stage** and **consider which process activities can be outsourced** versus **which must remain with Bankolino**.

This means that **these applications do not have to be further developed and operated**, and can even **be removed if the BPO partner brings its own application landscape**.



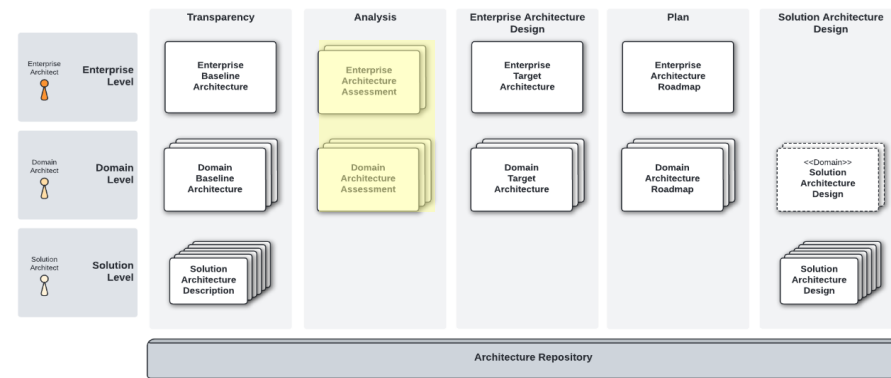
However, it may be necessary to **significantly adapt** or even **completely redevelop** the existing application **"HireReview"** to **integrate the relevant capabilities** into a **completely new application landscape** operated by the BPO partner.

Case Study

Overview

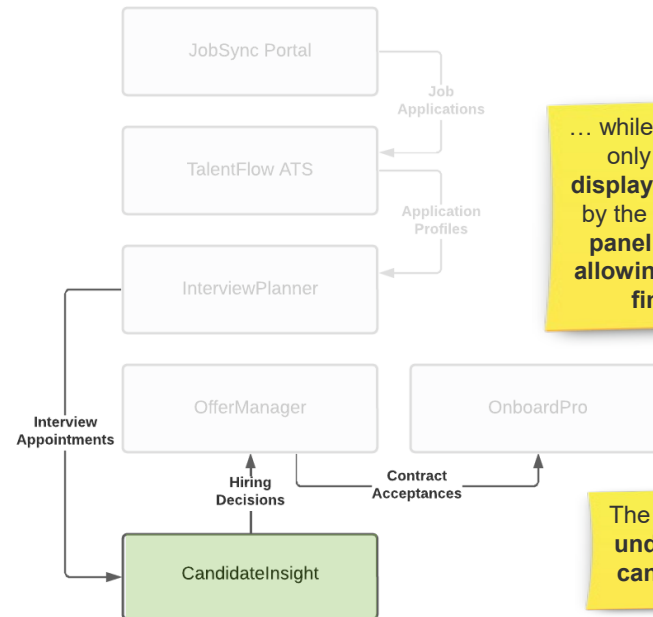
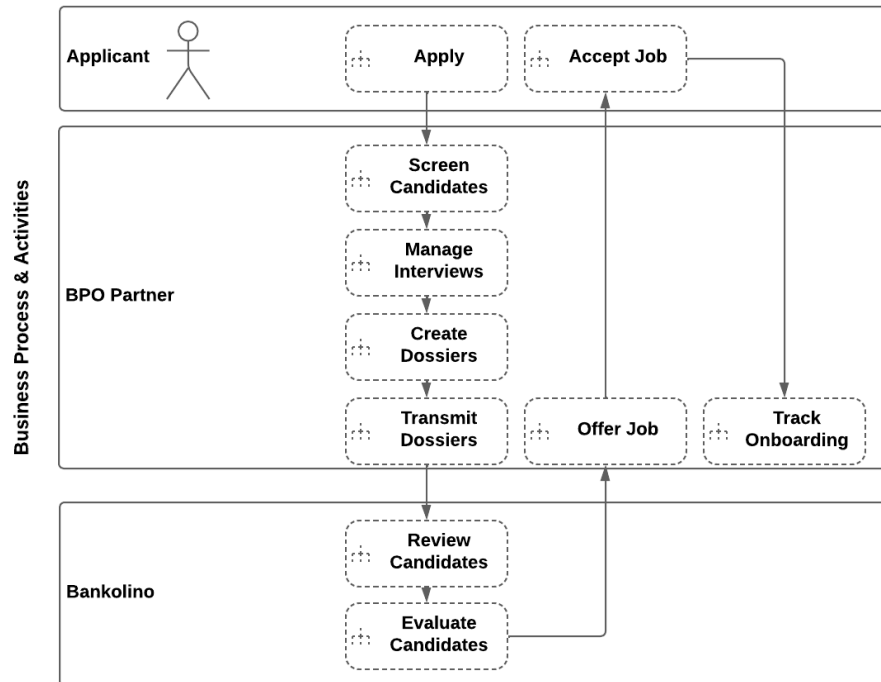
What **options** do you see for **BPO** in the **situation described**, and **how** would you **benefit** from this?

Case Study BPO Example



The Enterprise and Domain **Architects** currently **assume** that a **new application** for the **internal hiring panel** needs to be **established**. They have given it the **working title** “**CandidateInsight**”.

The **plan** is that **all tools** (except **from CandidateInsight**) will be **provided** by the **future BPO partner** ...



... while **CandidateInsight** remains the only **Bankolino-internal tool** that **displays applicant dossiers** provided by the **BPO** for the **Bankolino hiring panel** in an **interactive dashboard**, allowing **comments, evaluations** and **final decisions** to be made.

The **system** will remain **entirely under Bankolino's control** and **cannot be viewed** by the **BPO**.

Case Study Overview

Who has ever **pursued** a **similar option** in **your own professional life**?

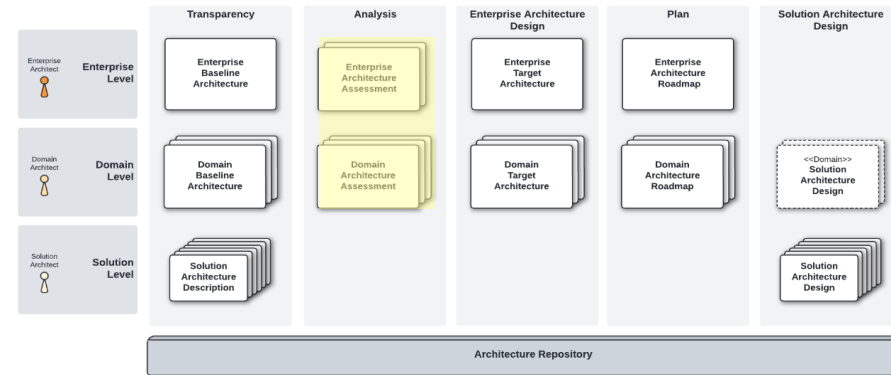
0

I have already replaced applications
with BPO.

0

I have not yet replaced applications
with BPO.

Case Study BPO Example



The **Architects** draw an interim conclusion by **compiling the advantages and disadvantages** of a **BPO-based realignment** of the **recruitment process** (supported by the **Bankolino recruitment** department).

Pro and Con Assessment

Advantages

- Cost reduction through economies of scale of BPO.
- Access to modern recruitment technologies.
- Faster application process.
- Focus on more strategic HR issues internally.

Disadvantages

- Loss of control over applicant pre-selection.
- Dependence on external partners.
- Risks regarding data protection and compliance.
- Loss of internal expertise.

The **Architects** conclude that, **bottom line, strategic control** is **maintained by outsourcing only the operational process**, while **key decisions** (e.g., hiring panel **via CandidateInsight**) **remain internal**.

Case Study

Overview

How do **development costs** compare to **operating costs** of the **total cost of ownership (TCO)** for **any** given **application**?



Case Study BPO Example



Next, the **Architects** examine which **companies** offer the required **Recruitment and Talent Acquisition Services** in a **BPO model** on the market.

TalentHive AG

Service Model

Provides its own services and applications.

Services

- End-to-end recruitment platform including AI-based screening.
- Own applicant portal, interview scheduling, offer preparation, onboarding.
- API-based transfer of pre-selected candidate files to Bankolino

Integration Approach

Only specific transfer points, no operational use of Bankolino systems necessary.

RecruitPro Services GmbH

Service Model

Uses the Client's own applications.

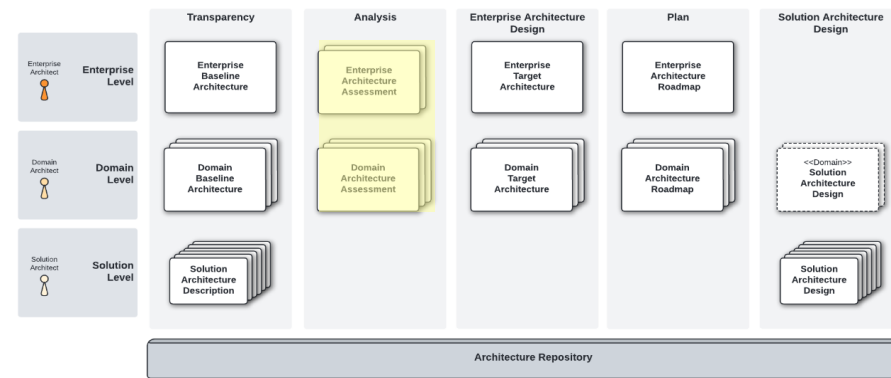
Services

- Advertising job vacancies via Bankolino systems.
- Processing applications in TalentFlow ATS.
- Planning interviews in InterviewPlanner.

Integration Approach

RecruitPro employees work as an 'extended workforce' directly in Bankolino's IT department.

After conducting **extensive research**, they **select two potential BPO candidates**, which become subject to a more **detailed review and final assessment**.

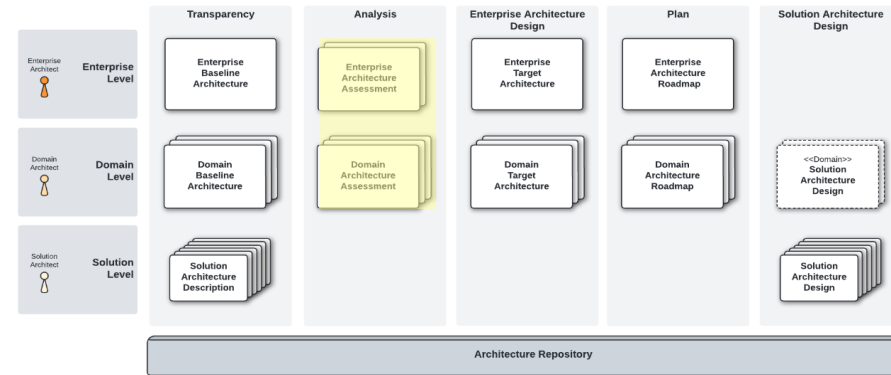


Case Study Overview

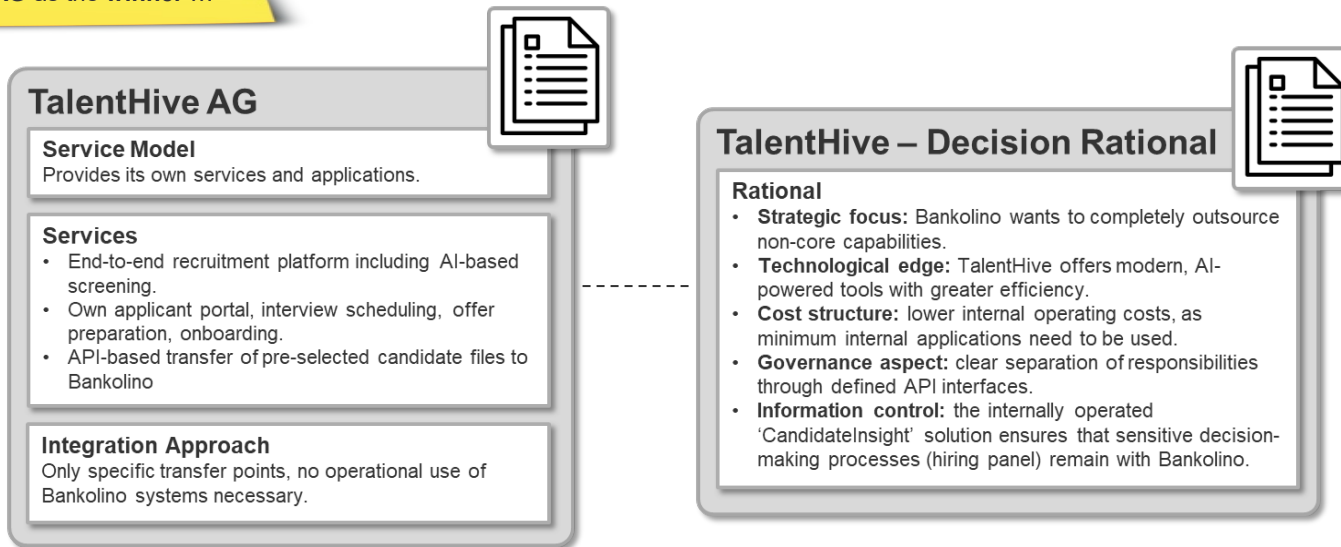
Which of the **2 companies** would you choose as your **BPO partner**?



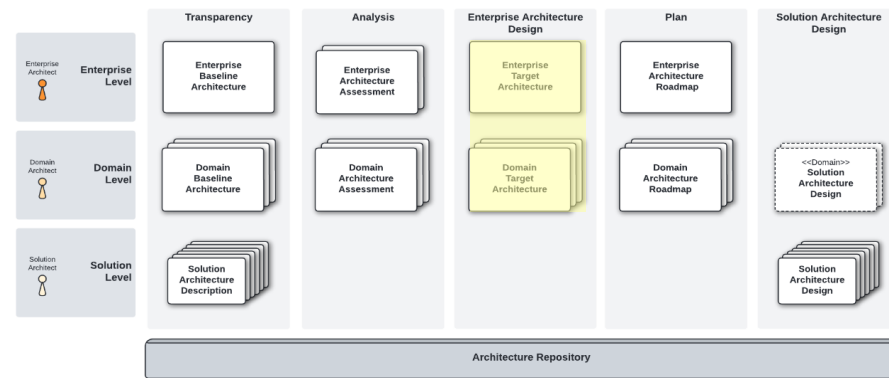
Case Study BPO Example



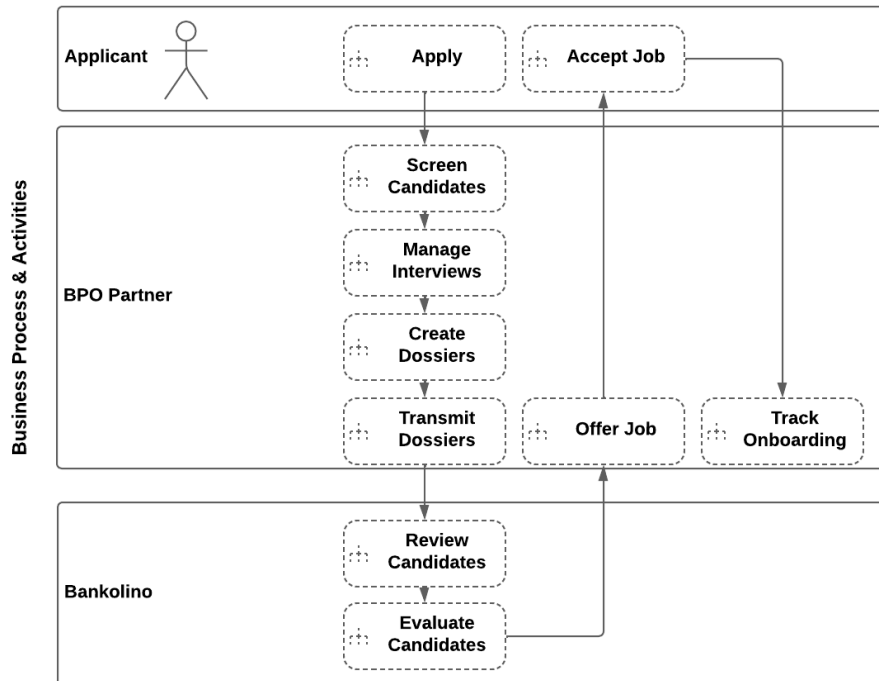
The detailed assessment reveals
TalentHive AG as the winner ...



Case Study BPO Example

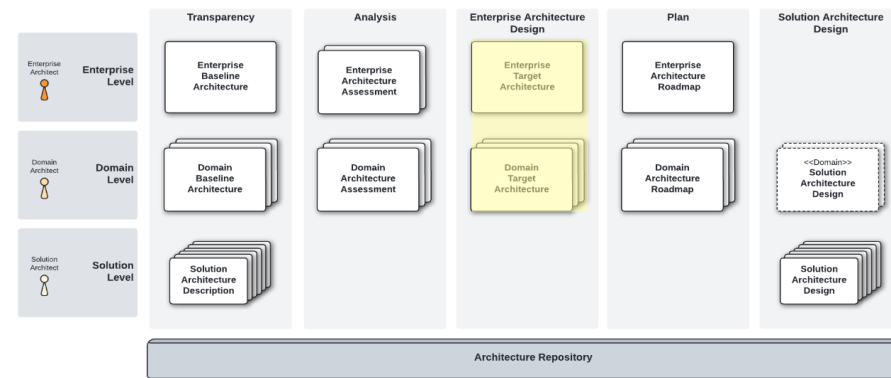


The **Bankolino Enterprise** and the responsible **Domain Architect** elaborate a future (high-level) **process design** for the **Recruitment and Talent Acquisition Process**, which describes the situation after business process outsourcing.

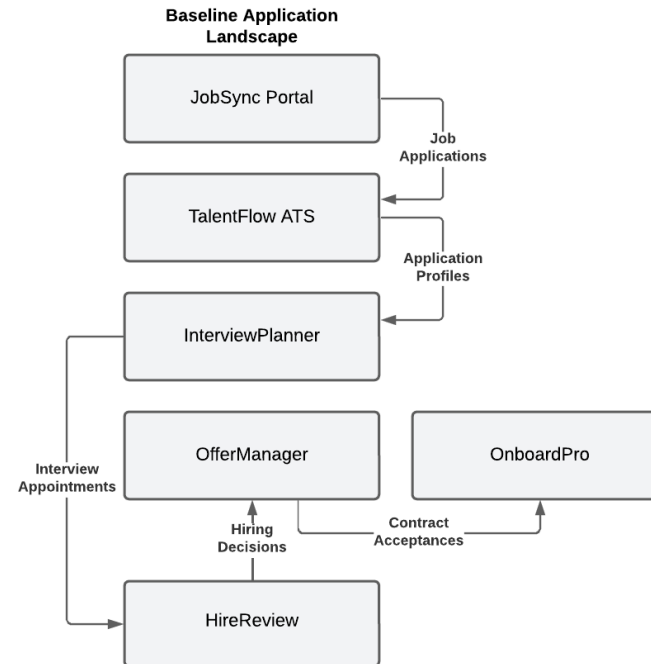
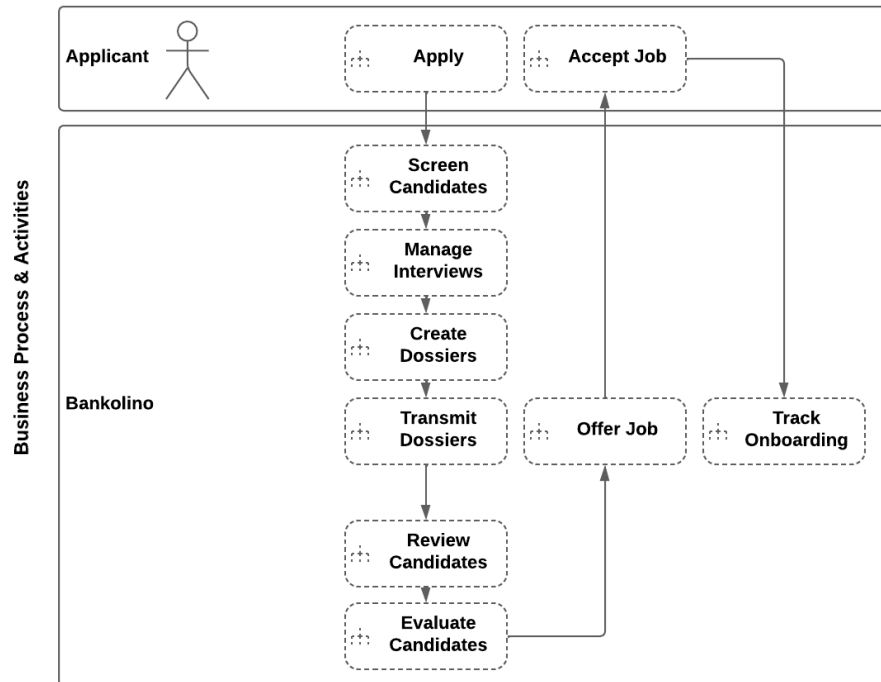


The **process** clearly shows that the **majority of recruitment and talent management activities** in the future process **will be handled** by the **BPO partner**, with **only candidate review and evaluation** remaining with **Bankolino**.

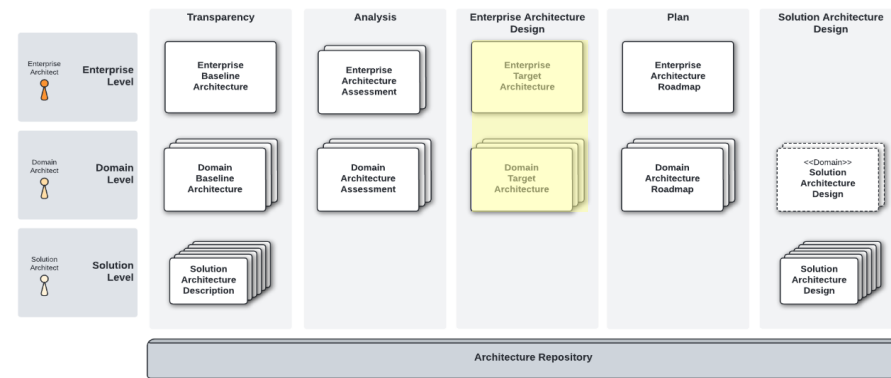
Case Study BPO Example



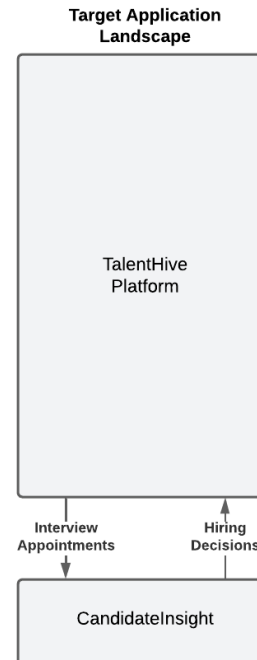
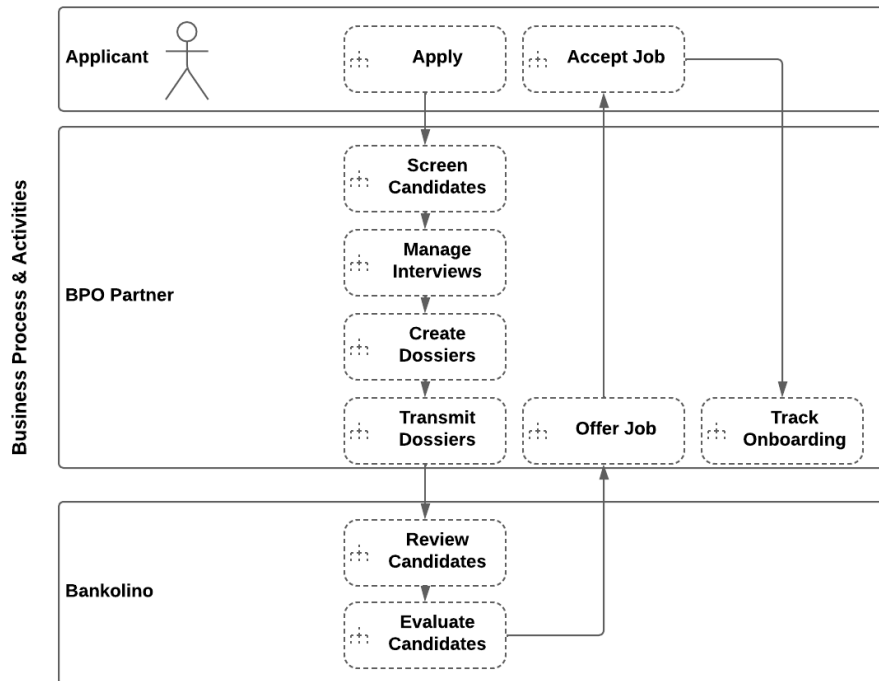
The **current situation (Baseline Application Landscape)**, in which **Bankolino evolves and operates several own applications**, will be **transferred** as part of the **BPO conversion ...**



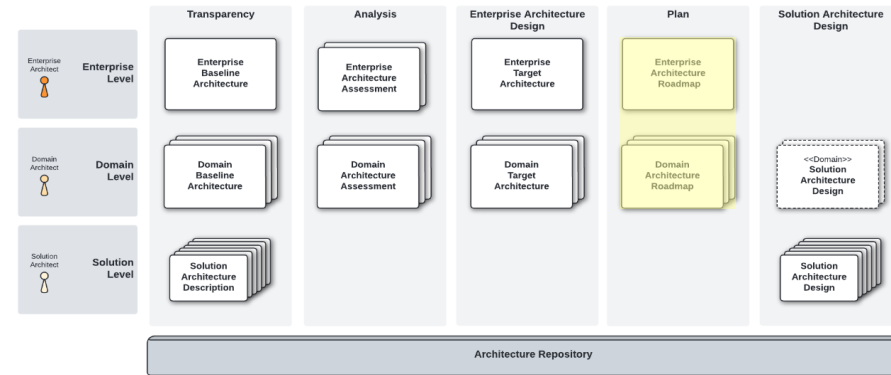
Case Study BPO Example



... to a situation (Target Application Landscape) in which the future BPO partner TalentHive will support its own services with its own applications, while Bankolino will only remain responsible for one application (→ CandidateInsight).



Case Study BPO Example



In the **next step**, the **Enterprise** and **Domain Architect** compile a **Project Charter** for implementation planning based on the information currently available.

Project Proposal – BPOing Recruitment & Talent Acquisition



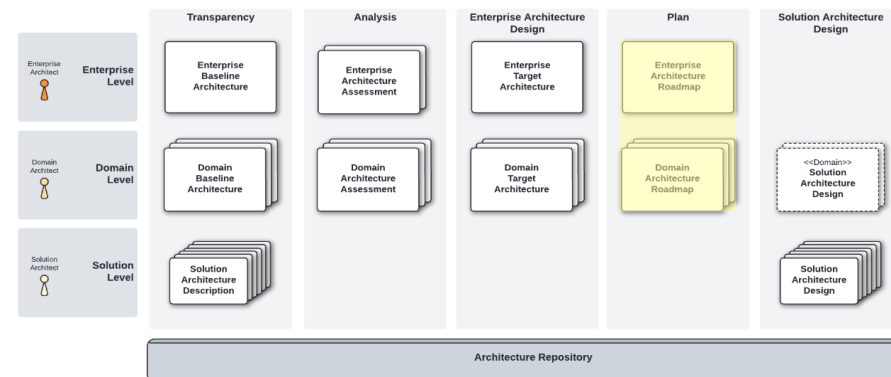
Case Study

Overview

What are common ingredients of a **Project Charter**?

- 0 Objectives
- 0 Scope
- 0 Timelines
- 0 Resources
- 0 Budget
- 0 Risks
- 0 Dependencies
- 0 Deliverables
- 0 Assumptions
- 0 Prerequisites
- 0 Consequences of not doing it

Case Study BPO Example



In the **next step**, the **Enterprise** and **Domain Architect** compile a **Project Charter** for **implementation planning** based on the **information currently available**.

Project Proposal – BPOing Recruitment & Talent Acquisition

Purpose

Bankolino intends to outsource the operational components of its recruitment and talent acquisition process to a specialised BPO partner (TalentHive AG). The aim is to make non-strategic HR functions more efficient and cost-effective without relinquishing the final decision-making authority.

Objectives

- Establishment of an efficient BPO-driven recruitment process.
- Replacement of existing HR applications in the recruitment area.
- Development and implementation of the internal application 'CandidateInsight'.
- Ensuring a smooth integration and transition process

Deliverables

- Integration architecture and interface design.
- Implementation and rollout of the CandidateInsight solution at Bankolino.
- Decommissioning of obsolete systems at Bankolino.
- SLA-based operating agreement with TalentHive.
- Training for hiring panel and support staff

Assumptions

- TalentHive complies with regulatory requirements.
- Internal hiring panel remains consistent in structure and scope.

Scope

In Scope

- All applications of the recruitment process.
- Data migration as necessary.
- Interface and security concept

Out of Scope

- Payroll or other HR processes.
- Personnel development & retention management

Timelines

Phase	Timeline	Deliverables
Initiation	Q1 2026	Project organisation, contract with TalentHive
Architecture	Q2 2026	EA specification, API design
Implement.	Q3 2026	CandidateInsight prototype & Testing
Migration	Q4 2026	Rollout of TalentHive, Training
Go-Live	Q1 2027	Setting System productive.

Risks

- Interface incompatibility.
- Resistance from internal stakeholders.
- Data protection & compliance risks.

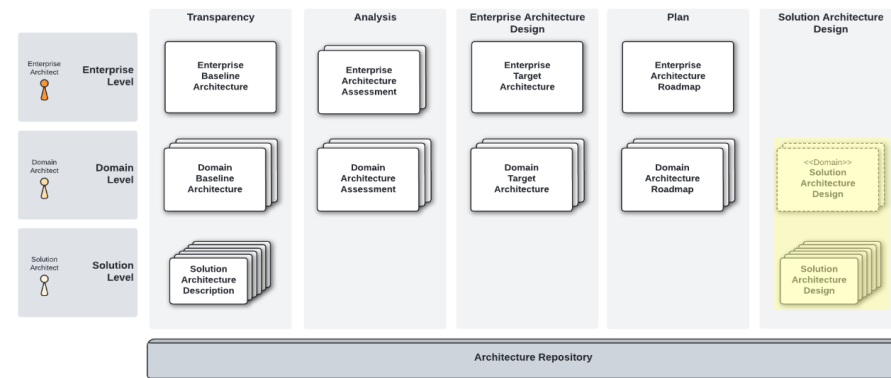
The “**BPOing Recruitment & Talent Acquisition**” project is **applying** for the **necessary resources** (e.g., budget, people) as part of **Bankolino's project portfolio management** and will be **launched once approved**.

Case Study BPO Example

Domain Architect

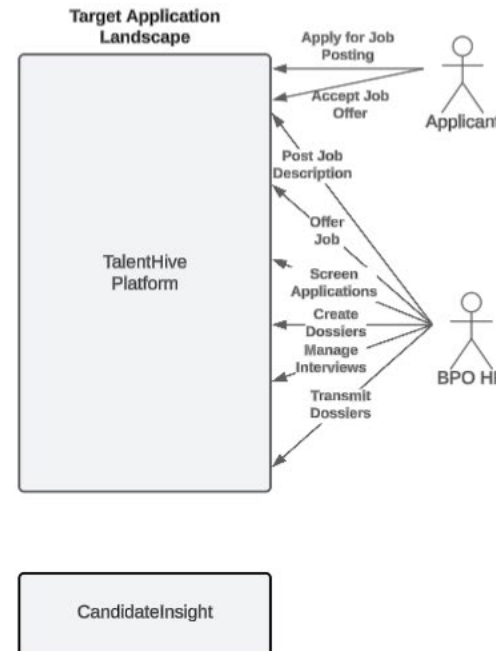
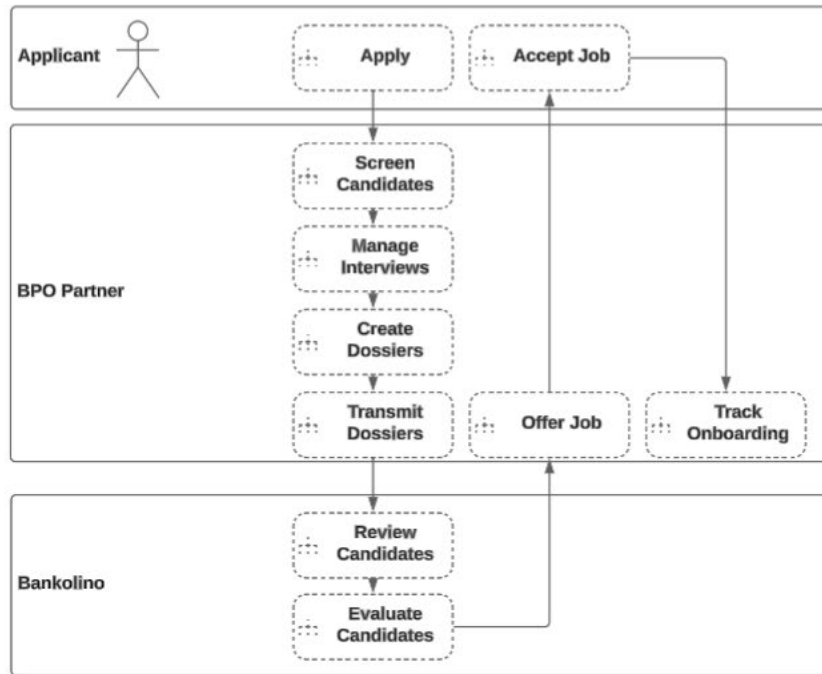


Solution Architect



Domain and solution architects initially reflect on the existing plan together and then zoom in on the details.

Business Process & Activities



Case Study

Overview

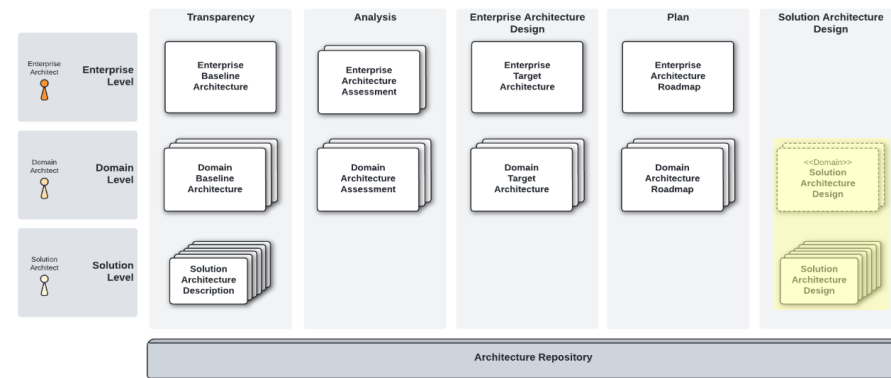
Why are the **functionalities** of the **TalentHive platform** also **identified** even if they are the **responsibility of the BPO partner**?

Case Study BPO Example

Domain Architect

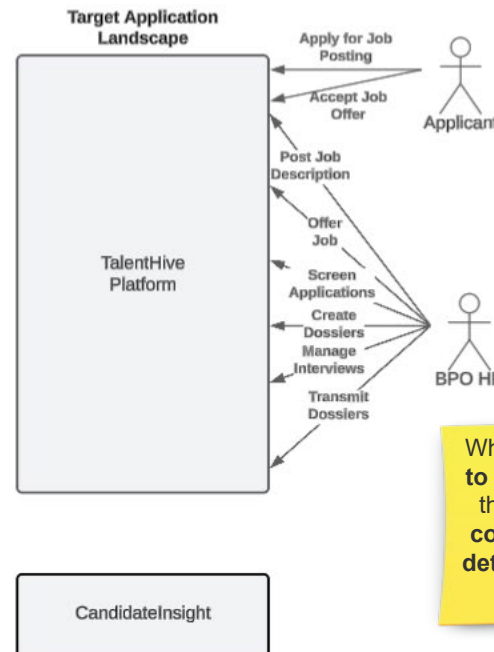
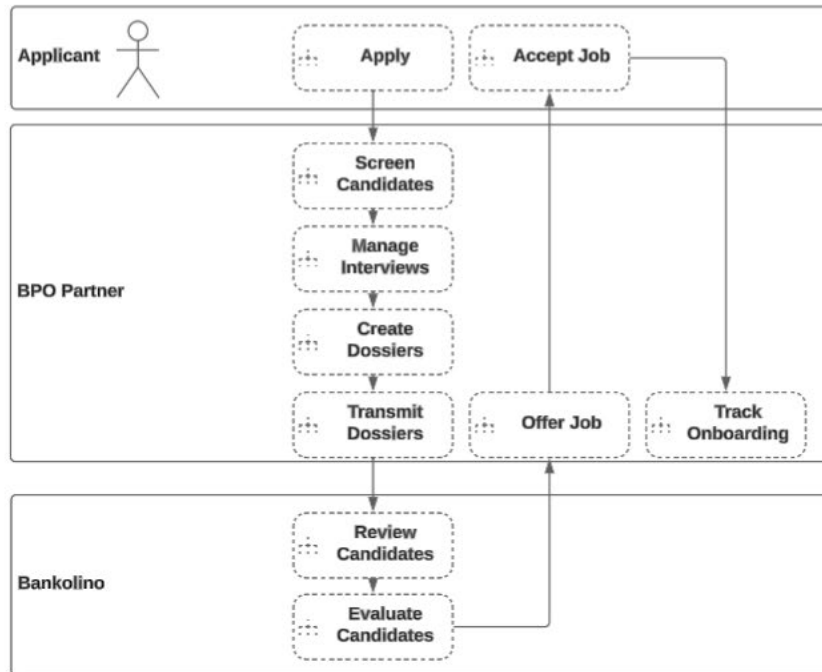


Solution Architect



Domain and solution architects initially reflect on the existing plan together and then zoom in on the details.

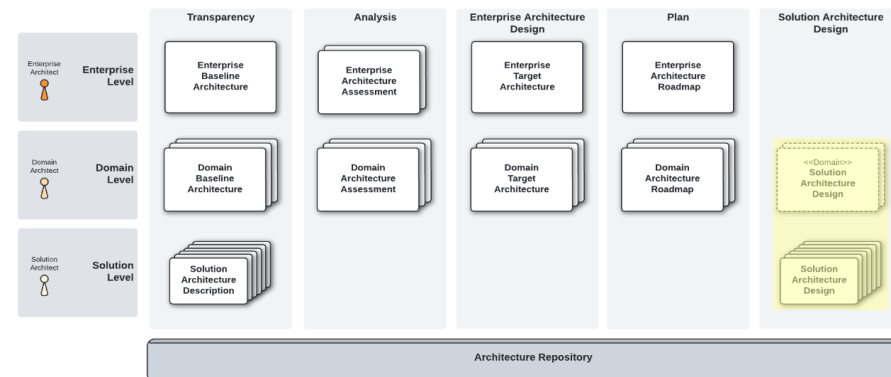
Business Process & Activities



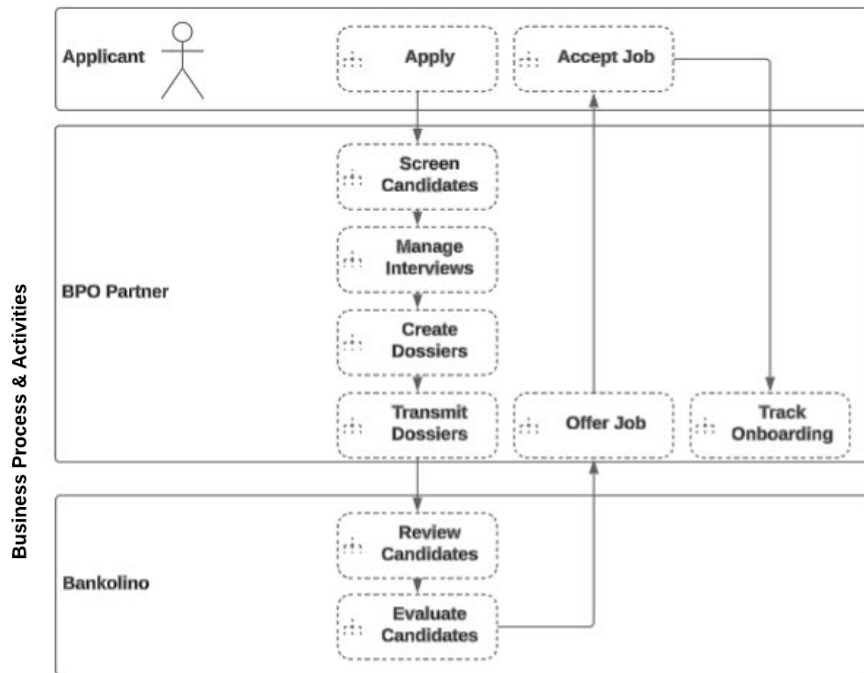
In a first step, they **identify** the basic **functionalities** required on the **TalentHive** platform.

While these **capabilities** do **not** have to be developed by **Bankolino** itself, they **must** be included in the **BPO contract** and **specified** in **sufficient detail** as **functions** to be provided to **Job applicants**.

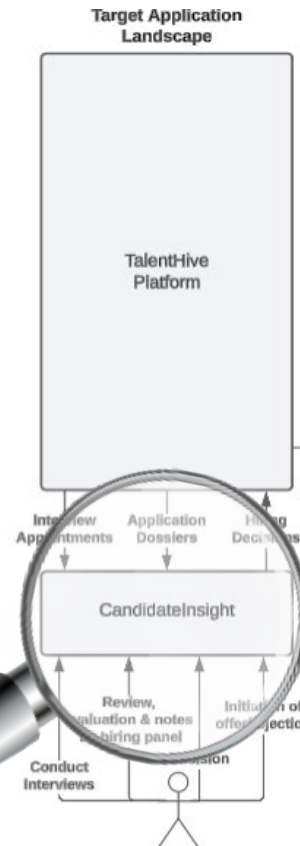
Case Study BPO Example



Domain and solution architects initially reflect on the existing plan together and then zoom in on the details.



Next the Solution Architect zooms further in on the CandidateInsight Application.



In the next step, they identify the basic functions that CandidateInsight must offer its users (i.e., members of the hiring board) in Bankolino.

Further, they identify the information that will need to be exchanged between CandidateInsight and the TalentHive platform in the future.

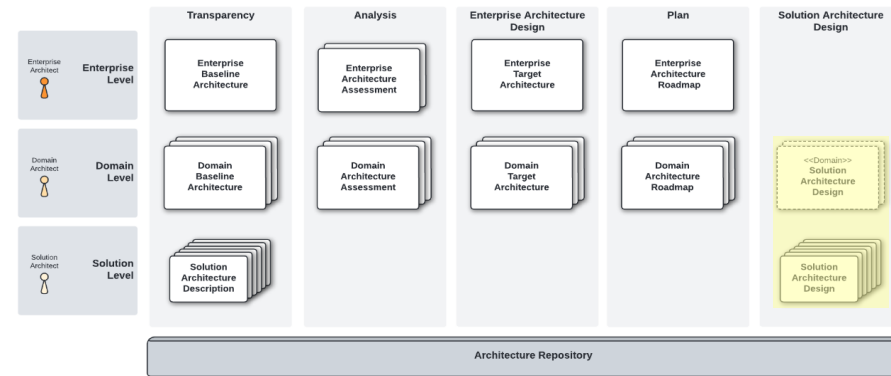
In addition, other Bankolino applications are identified, that exist in the wider environment and also need to be connected.

Case Study BPO Example

Domain
Architect



Solution
Architect

In a first step, a **workshop** is held with potential **hiring board members** and other **individuals familiar with the domain**.

Expectations for the new **CandidateInsight** application are **discussed, formulated, and captured** in a **preliminary application factsheet**.

CandidateInsight

Key Capabilities

Case Study Overview

Next, we will **divide into six working groups** again.

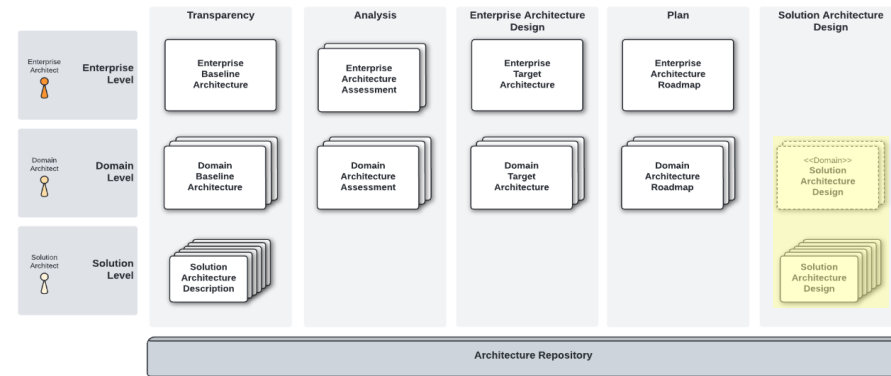
Each group will **collect** and note any **additional functional** and **non-functional capabilities** that are **expected of CandidateInsight** in the future so that this new application can be fully integrated into the scenario outlined so far.

Each group goes through the following **three steps**:

- **Identify** and **determine** the **categories** in which you suspect relevant **requirements** exist (e.g., security).
- **Collect** and **describe concrete requirements for CandidateInsight** that you consider relevant.
- **For each** of the **requirements** you have identified, name the responsible **stakeholder group** with whom you will confirm the requirement as well as discuss it in more detail.

Collect your results and then **present** them to the whole **class**.

Case Study BPO Example



In a first step, a **workshop** is held with potential **hiring board members** and other **individuals** familiar with the domain.

Expectations for the new **CandidateInsight** application are **discussed**, **formulated**, and **captured** in a **preliminary application factsheet**.

As we can already see from this **very rough first step**, there is a **wide range** of very **different types of aspects** that **need** to be **addressed** by the **new solution**.

Inter-system integration and synchronization aspects.

UI and usability aspects.

Identity and Access Management aspects.

CandidateInsight

Key Capabilities

Receipt and presentation of applicant dossiers

- Automated import via API from the BPO partner.
- Visualisation of structured applicant information and documents (CV, evaluations, screening results).

Evaluation and commenting by the hiring panel

- Role-based access control for panel members.
- Evaluation matrix (e.g. score, traffic light rating, text comments) Option for discussion and collaborative commenting.

Decision-making

- Consolidation of individual evaluations.
- Recording of the final hiring decision.
- Transfer of the decision to the HR system or to the BPO for further processing

Audit & compliance

- Logging of all evaluations, comments and decisions.
- Traceability of decision-making processes (auditability).

Security

- Access only for authorised employees.
- Encryption of sensitive personal data.
- GDPR-compliant data processing.

Core Use Cases.

Non-Repudiation aspects.

IT-Security aspects.

Data Protection and Privacy aspects.

Questions

